sAIfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics
Faces

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Outline of the talk

- **Faces**
  - What are they
  - Pre-processing
  - Feature extraction
  - Matching
- **Liveness Detection**
- **Deep fake**
  - A modern challenge
- **Template update**

Between physiological and behavioural biometrics

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- **Personal identity**
- **Emotional state**
- **Physiological state**

Pros and cons

- **Non-intrusive**
- **No cooperation from the user**
- **Multiple acquisitions**
  - Video-surveillance
- **Dealing with**
  - Lighting conditions
  - Pose and scale (distance from the camera)
  - Intra-class variations
  - Evasion
  - Presentation attacks

Intra-class variations and their causes

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- **Expressions**

- **Pose**

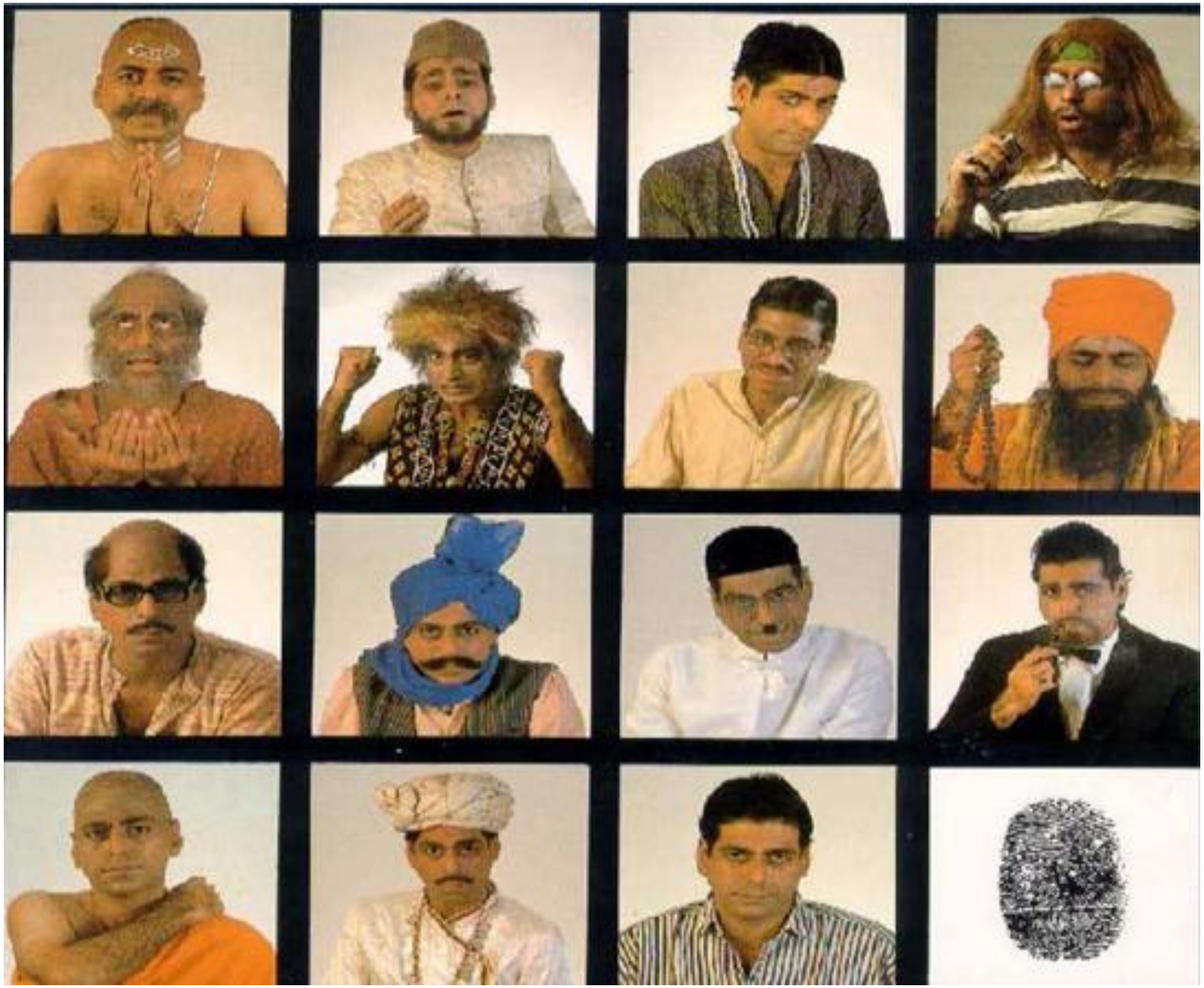
- **Lighting**

- **Aging**

Indoor and outdoor – any difference?

Relatives and twins

One face, N camouflages



Face capturing

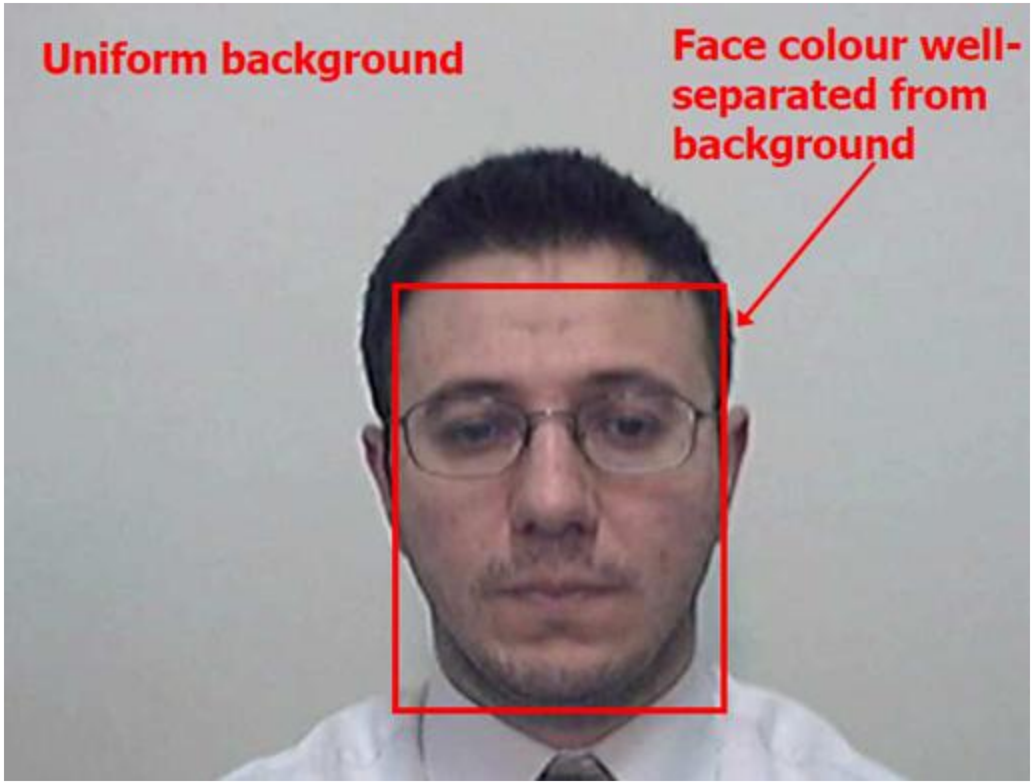
- **Still images**
  - Snapshots
- **Video-sequences**

Sensoring

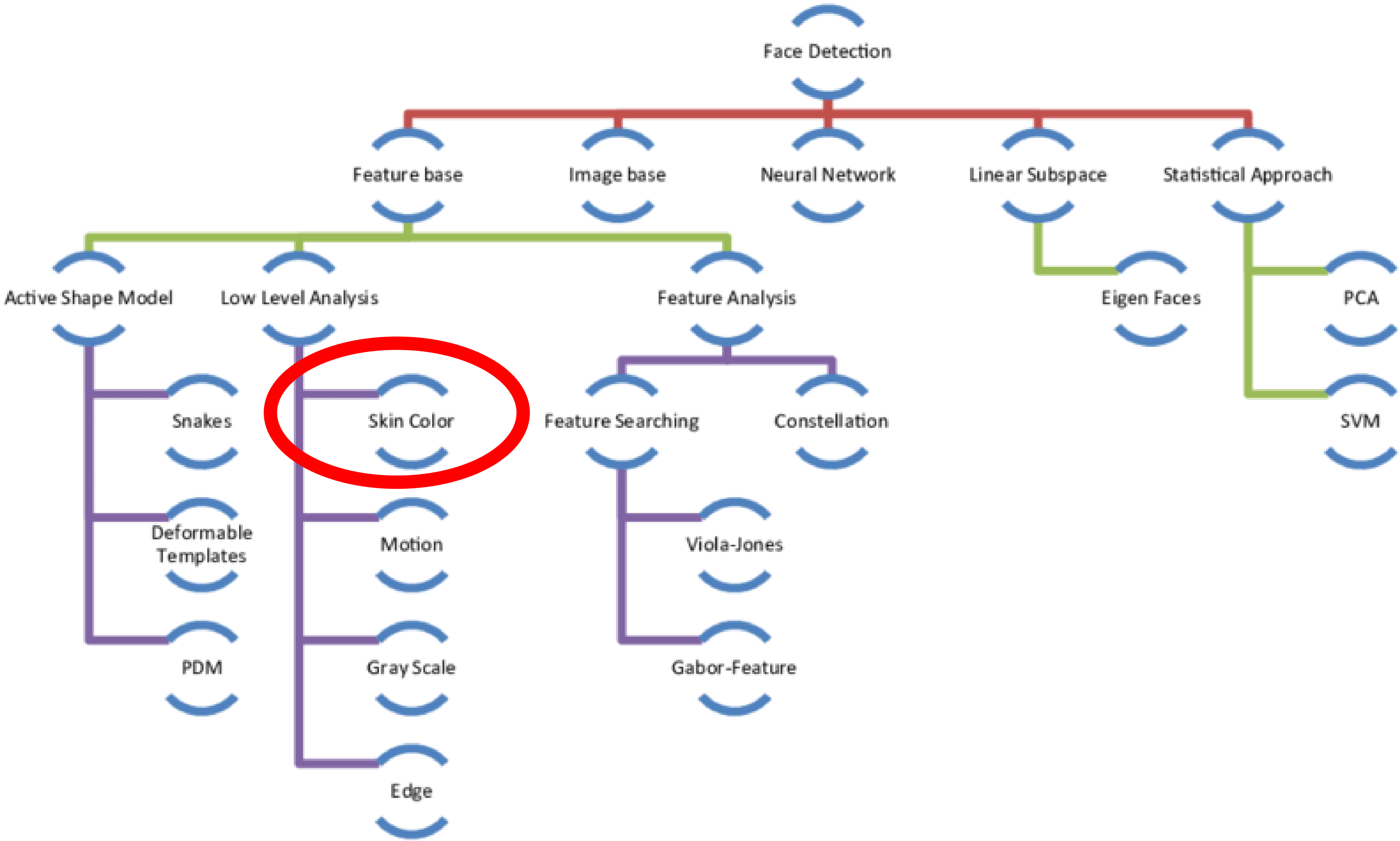
Controlled environment for face capturing

Pre-processing

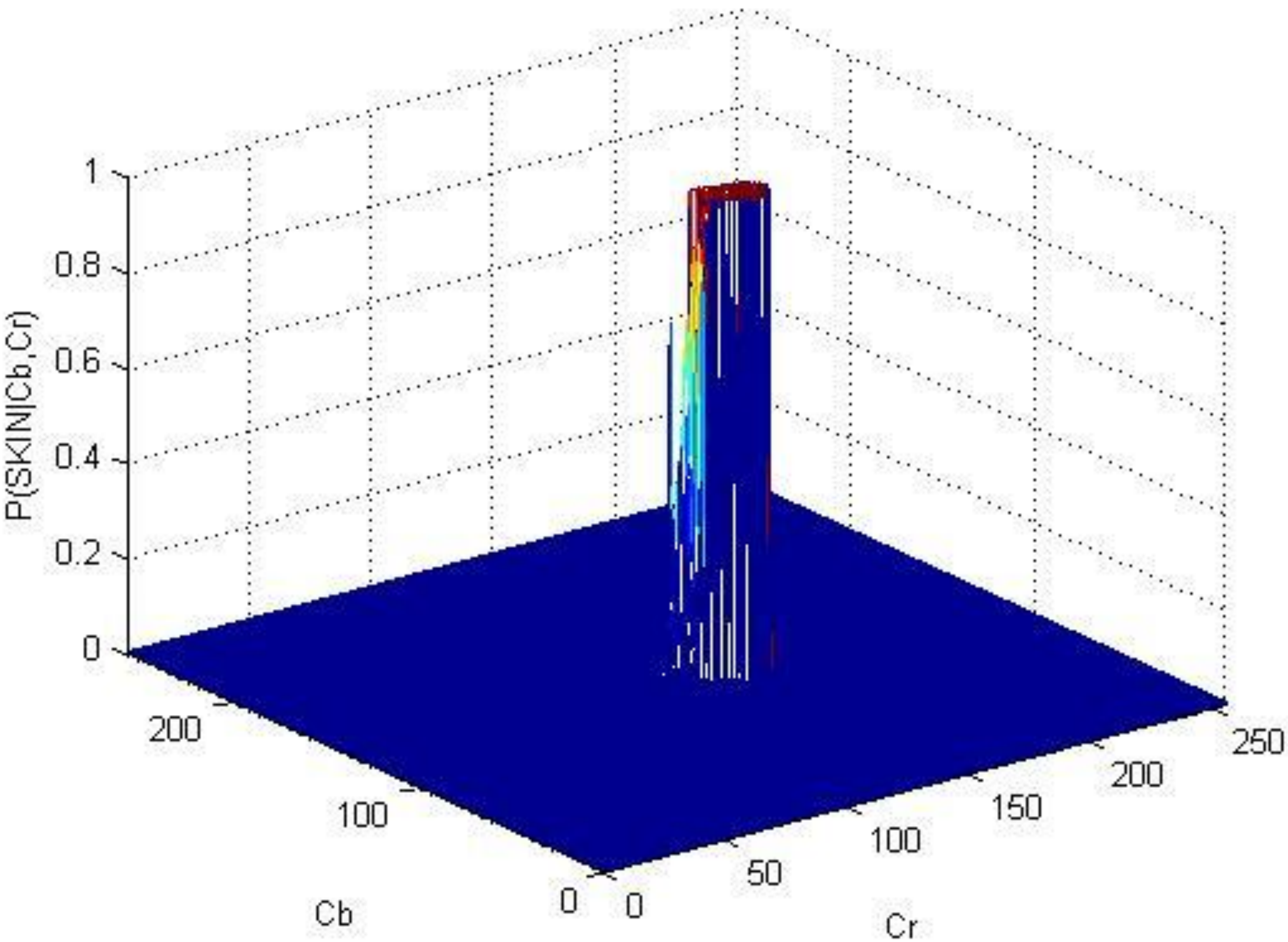
- **Face segmentation**
- **Face detection**
- **Face normalization**
  - «Canonical» face



Taxonomy of face detection approaches



Skin color-based face detection



Skin colour-based face detection

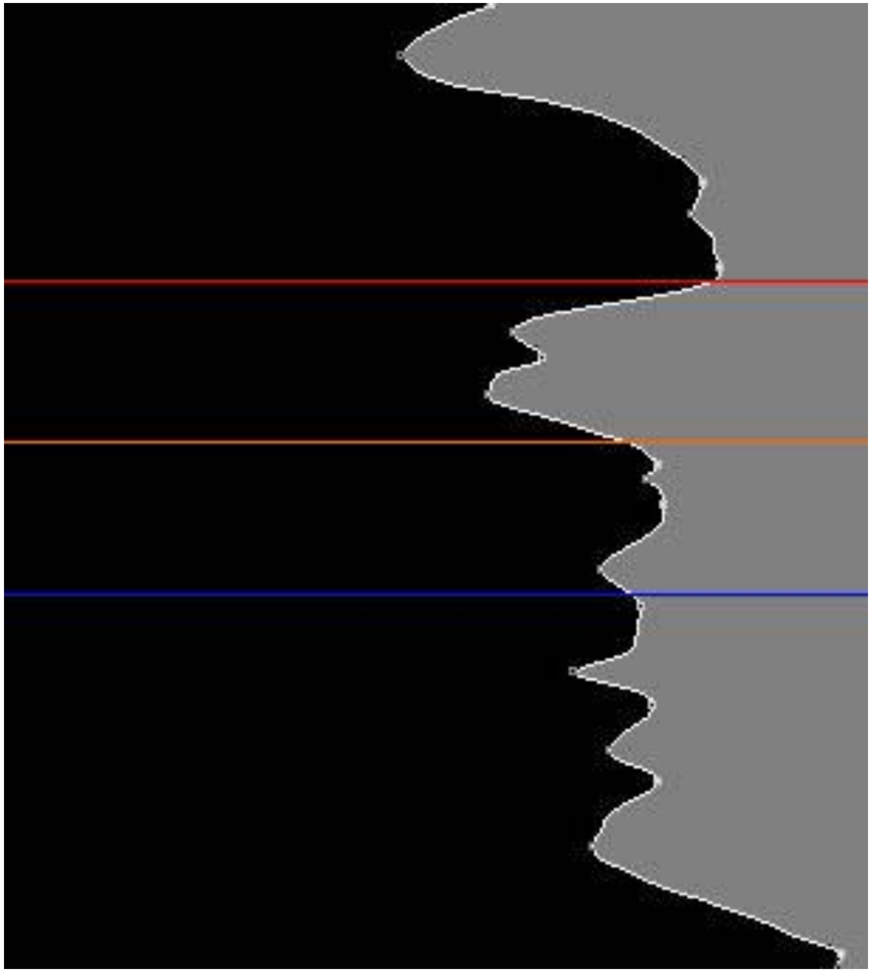
Homeworks

- **Write a Python (or C) function able to compute the P(SKIN|c) over a set of pixel samples in terms of crominance values c**
- **Write a Python (or C) function able to compute the skin map of a given image from the P(SKIN|x) distribution**
- **Write a Python (or C) function able to binarize a given facial images on the basis of the skin map**
- **Write a Python (or C) function able to implement an erosion morphological operator over a binarized image**
- **Design the skin color-based face detector on the basis of the functions above**

Be careful at…

- **Using representative pixel samples when estimating P(SKIN | c)**
- **Adopting an appropriate binarization threshold**

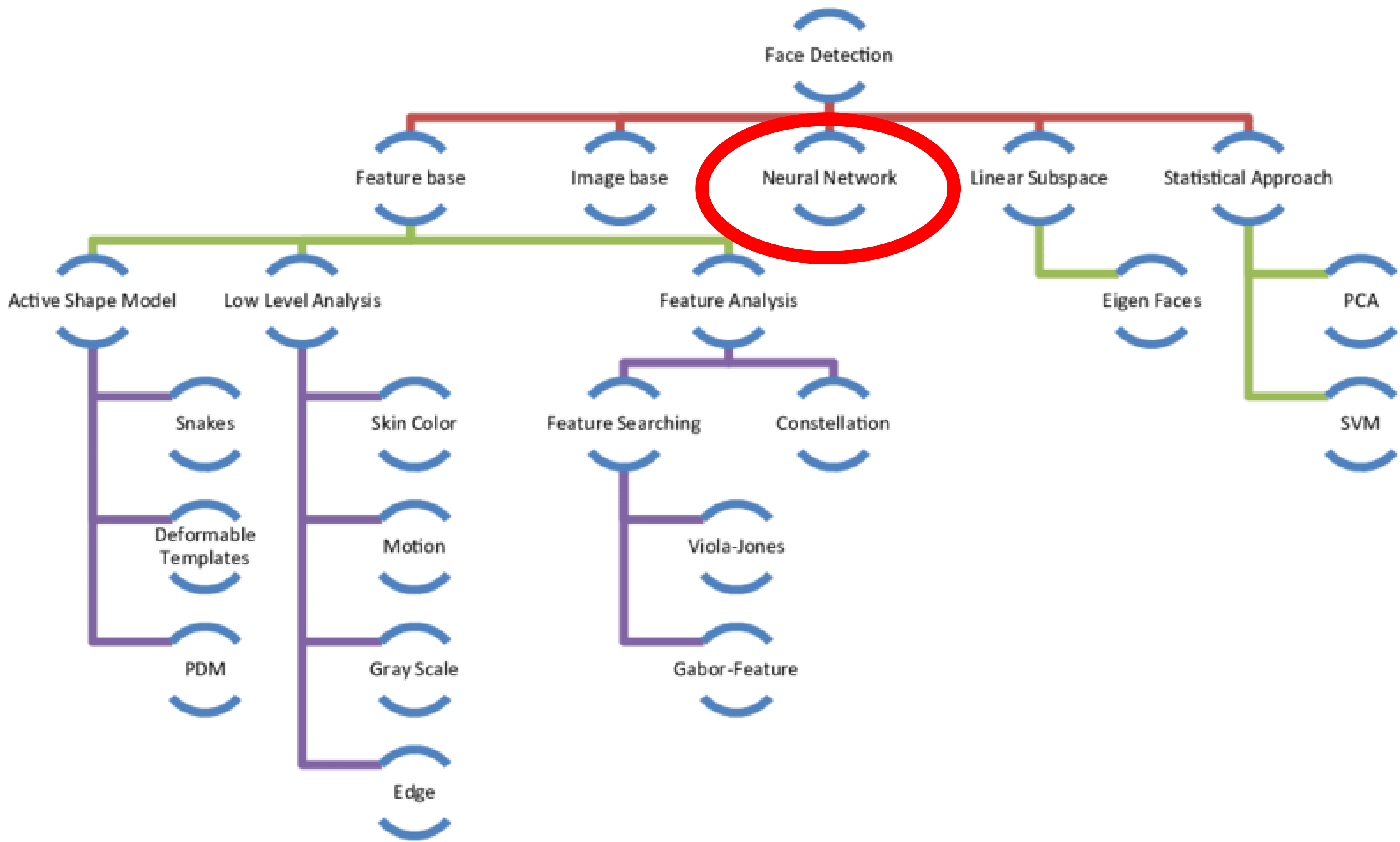
Face alignment by histograms projection



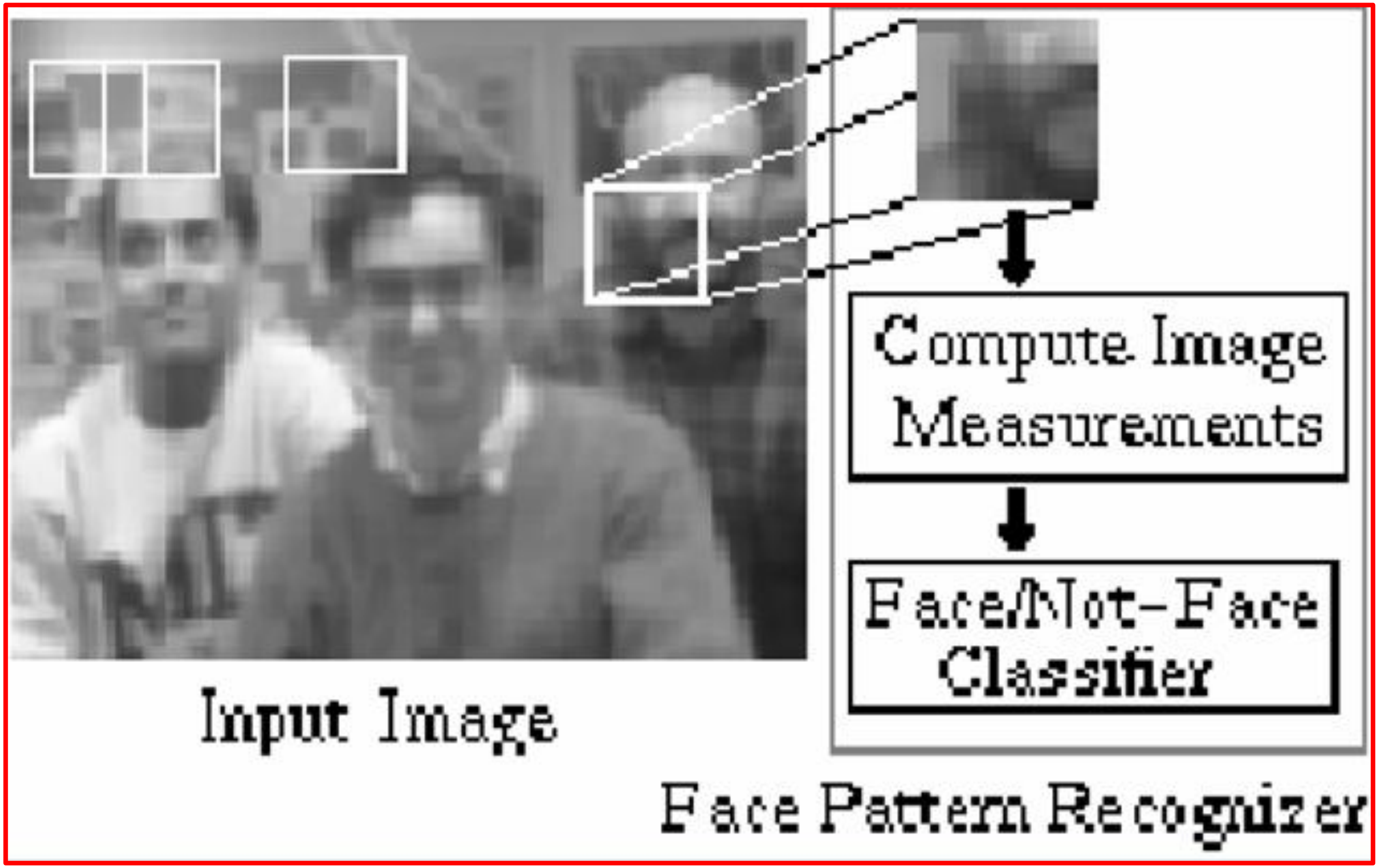
$$ipf_Y(Y) = \frac{1}{N_X} \sum_{x=1..N_X} img(y,x) \qquad ipf'_Y(Y) = \frac{ipf_Y(Y-1) + ipf_Y(Y) + ipf_Y(Y+1)}{3}$$

Canonical faces or face normalization

Taxonomy of face detection approaches



Neural network-based facial detection



Further pre-processing algorithms

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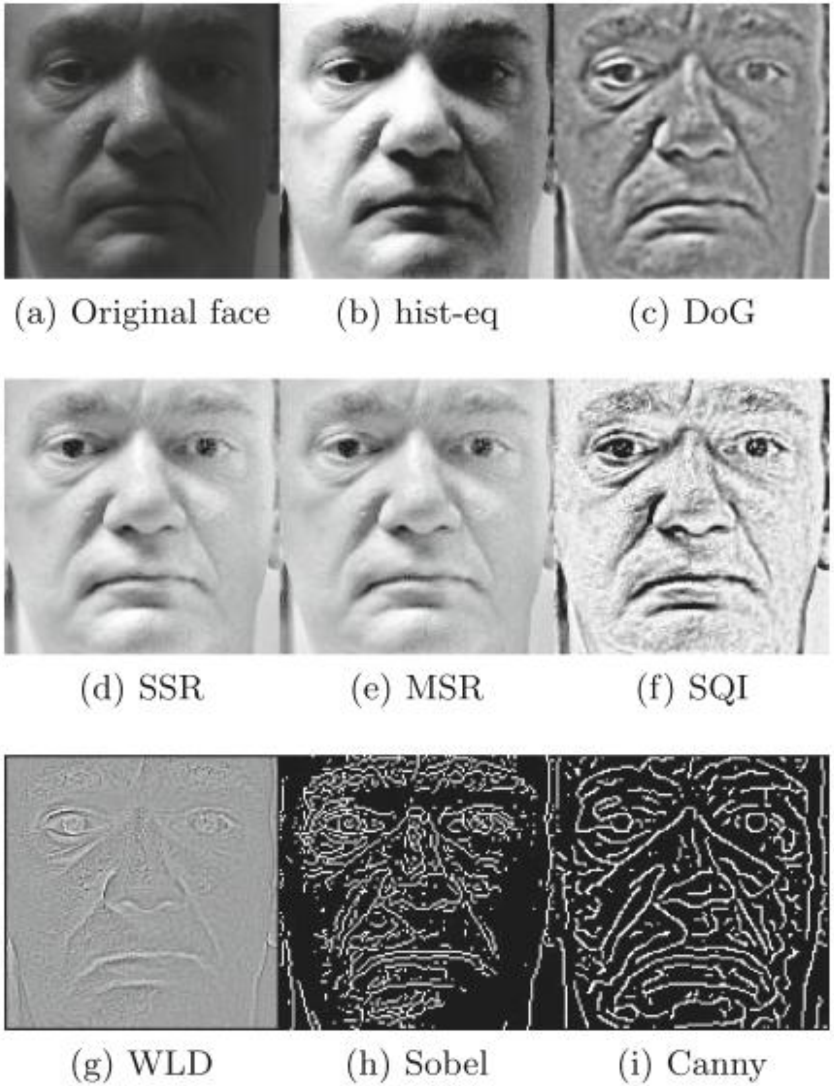


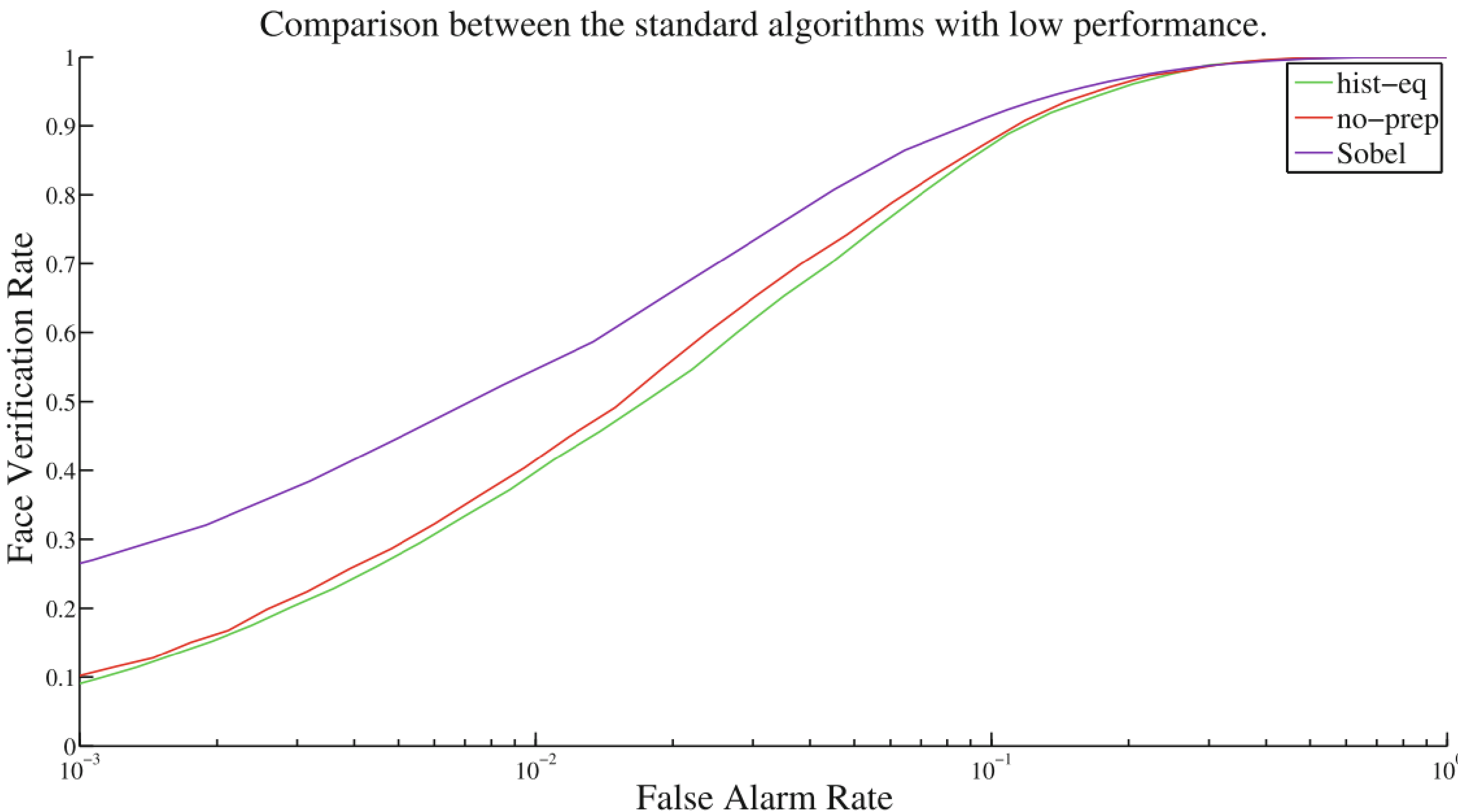
(a) Original face
(b) Histogram equalization
(c) Difference of Gaussians
- Noise removal by subtracting high and low frequency components

(d) Single Scale Retinex
(e) Multi Scale Retinex
- Logarithmic single and multiple bandwiths Gaussian filtered image

(f) Self Quotent Image
- Illumination estimation and subtraction

(g) Weber Local Descriptor
(h) Sobel filter
(i) Canny filter
- Edge enhancement

Pre-processing relevance

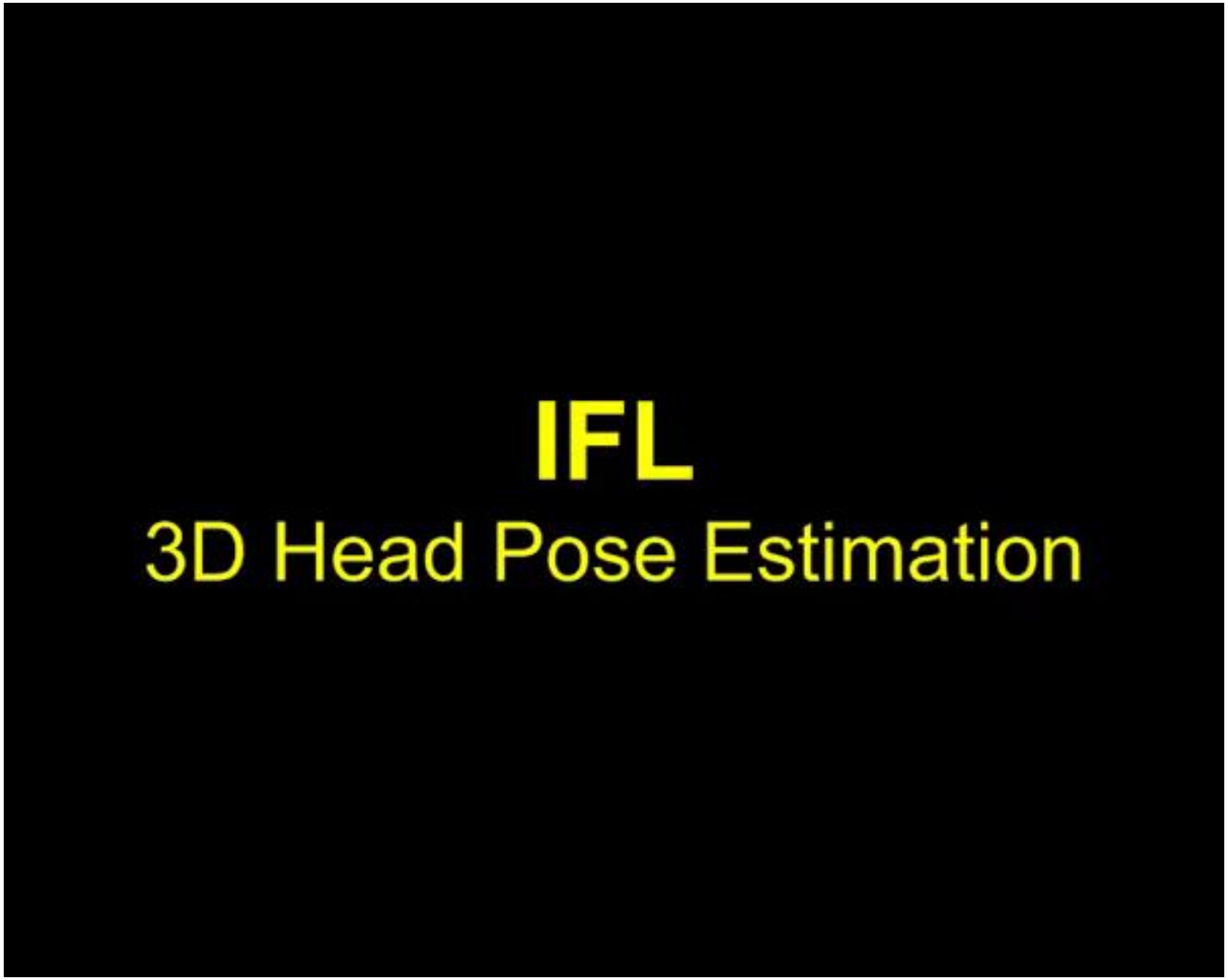


(c) Low performance face verification.

Exploring face quality

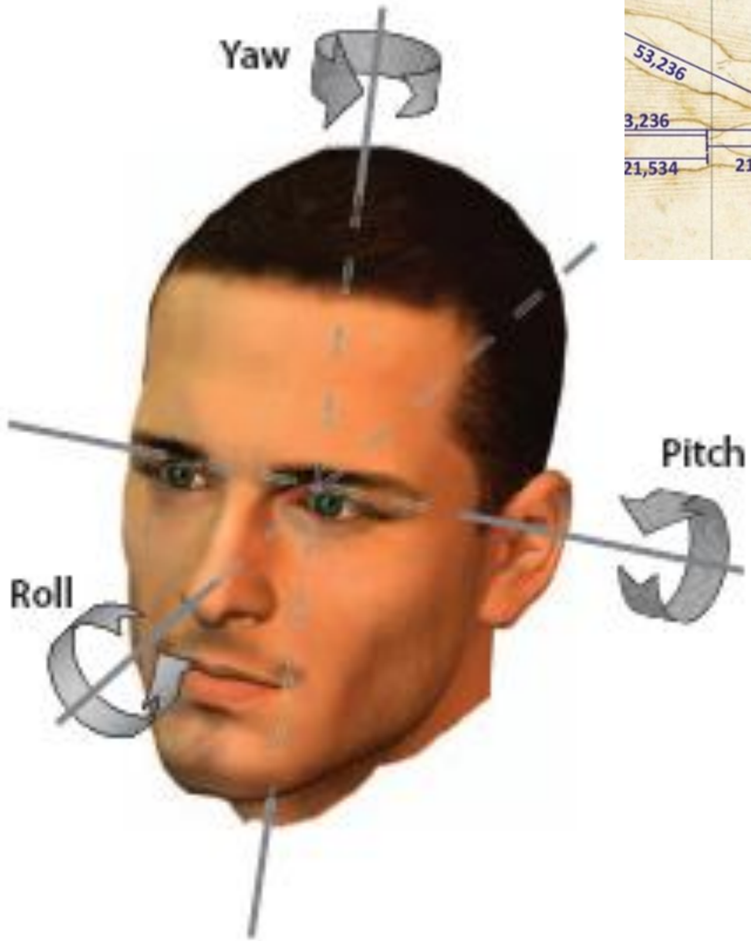
- **Lighting**
- **Blurring (focus)**
- **Pose estimation**
  - Best «frontal» face

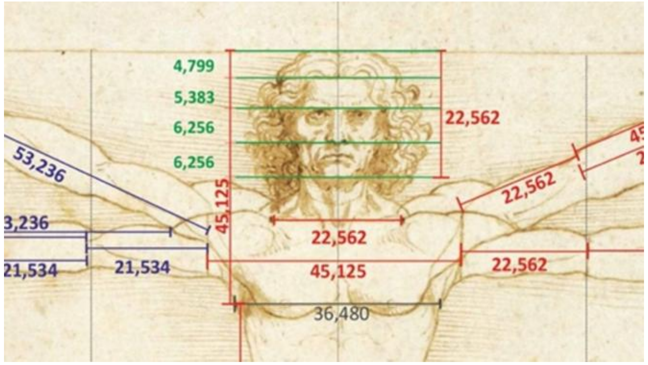
Pose estimation



- AdaBoost classifier for eyes detection
- Estimation of the nose area
- AdaBoost-based nose detection in that area
- Head pose estimation

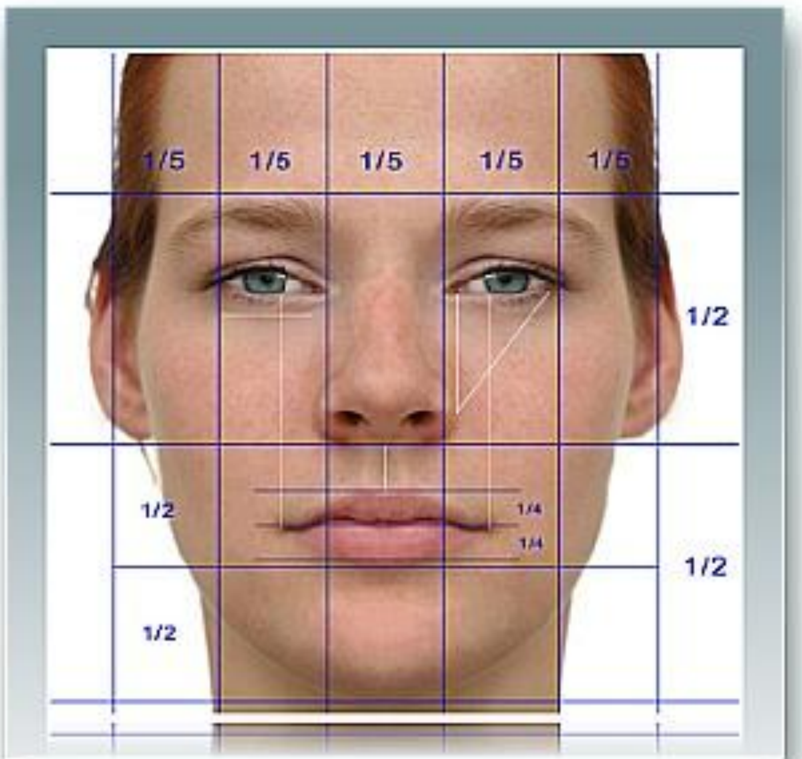
Head pose estimation



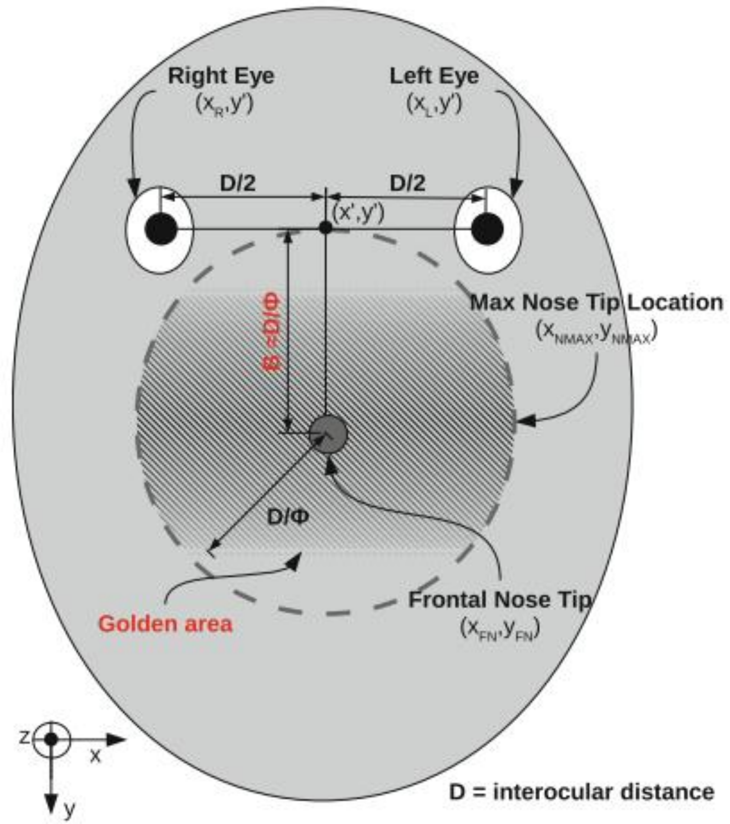


«Vitruvian man», Leonardo da Vinci, 1490

Nose area location

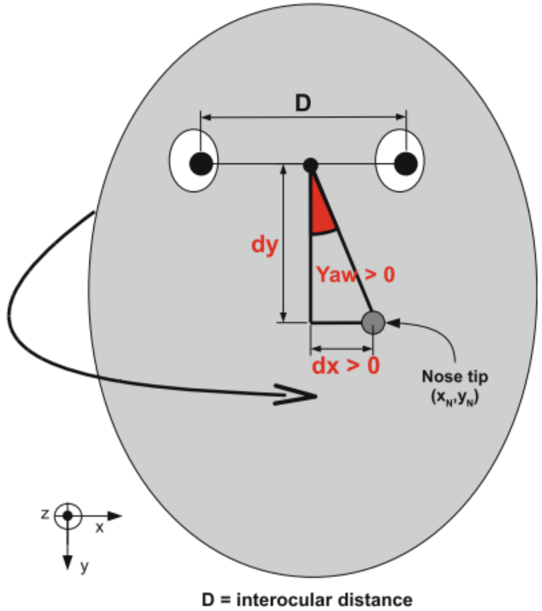


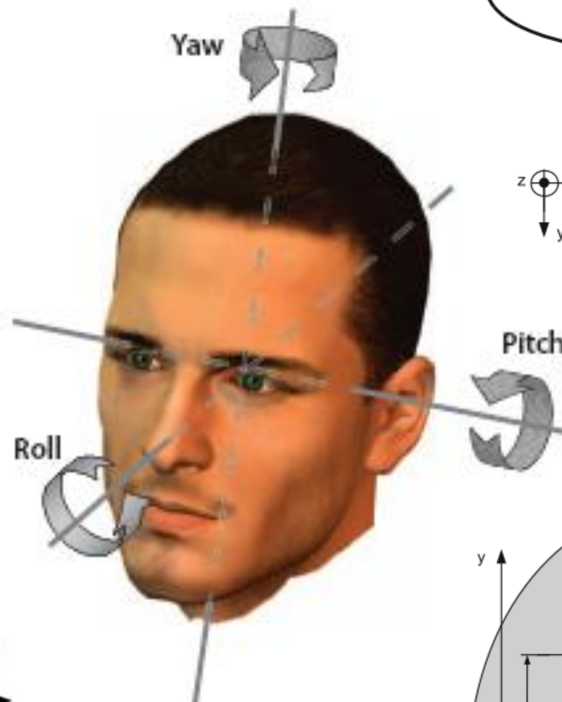
«Beauty» proportions
(plastic surgery)



«Beauty» proportions
(locations of eyes and nose)

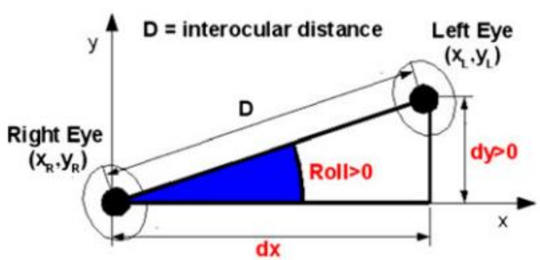
Yaw, roll, pitch angles



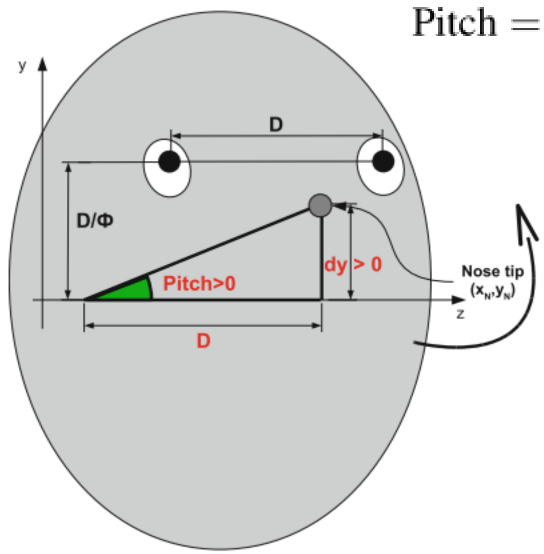


$$\text{Yaw} = arctg\left(\frac{dx}{dy}\right)$$

$$\text{Pitch} = arctg\left(\frac{dy}{D}\right)$$



$$\text{Roll} = arctg\left(\frac{dy}{dx}\right)$$



Face quality assessment

Feature-spaces for facial representation

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- **Appearance-based or holistic**
  - PCA, LDA, ICA, ...
- **Local feature-based**
  - Elastic bunch graph
  - Active shape models
- **Textural-based**
  - LBP, HOG, BSIF
- **Deep learning-based**
  - «Embeddings»

The problem of the feature space

VS

Can you see any discriminant features?

The problem of the feature space

VS

Can you see any discriminant features?

Principal components analysis

- **Let’s hypothesize having a face perfectly captured and normalized**

- **Problems**
  - Size of the feature space
  - A lot of redundant information
  - Potentially noisy
  - Highly correlated into a neighborood

- **Which are the «real» features?**

- **First possible solution**
  - Extracting highly uncorrelated features
  - Reducing the size of the feature space
  - Scalable solution (?)

The Karhunen-Loeve transform

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- **Let $X = \{x_1, .., x_N\}$ be a set of «vectorialized» facial images of size** $D = R \cdot C$
- **Let m be the related *sample mean***
- **The KL-transform compute the transformation matrix W such that, if the covariance matrix S defined as follows as follows:**

$$S = \sum_{i=1}^{n} (x_i - m) \cdot (x_i - m)^t$$

- **We have:**

$$\lambda e_i = S e_i$$

- **In other words, W is the eigenvectors matrix such that S is diagonalized**
- **The eigenvectors of W are called *eigenfaces***

Example- original set of images X- D=80x80

or.1

or.2

or.3

or.4

or.5

or.6

or.7

or.8

or.9

Example- eigenfaces ordered by eigenvalues

eigen.1

eigen.2

eigen.3

eigen.4

eigen.5

eigen.6

eigen.7

eigen.8

eigen.9

Residual faces

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- **Accordingly:**

$$\boldsymbol{y_i = W^t(x_i - m)}$$

- **By inverting the transformation, the *residual face* is** $\boldsymbol{x_i - m - W y_i}$

$$\boldsymbol{\|x_i - m - W y_i\|}$$

- **The value above is the amount of information lost during the process inversion**

- **This residual is the trade-off between the discriminant information aimed at representing each face and the noisy/redundant information we want to avoid by reducing the feature space**

Residual face

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- **The new co-variance matrix is <u>diagonal</u> ➔ meaning?**

- **We can select the first n eigenvectors such that, according to the *energy value* p:**

$$\boldsymbol{n} = \underset{i}{\operatorname{argmin}} \frac{\sum_{j=1}^{i} \boldsymbol{v}_j}{\sum_{j=1}^{D} \boldsymbol{v}_j} > \boldsymbol{p}$$

- **We can represent the same D-sized faces of X, as well the other faces in our «domain knowledge», by their correspondent transfomations of size $n \ll D$**

Example: $p = 0.95$; $n = 7$

recp.1

recp.2

recp.3

Do they look exactly the same as the original ones?

recp

recp.6

recp.7

recp.8

recp.9

Residual faces

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residual.1

residual.2

residual.3

residual.4

residual.5

residual.6

residual.7

residual.8

residual.9

Pros and cons

- **Pros**
  - Automatic feature extraction
  - No need of previous labelling
  - Very compact feature space
  - According to the Jacobi's method, if the number of images n is less than D, first n eigenfaces are computed and are sufficient to reconstruct all original «training» images

- **Cons**
  - Alignment is strongly necessary
  - Scalability problems
  - Never-seen-before faces

Linear Dependent Analysis

- **Same principle as PCA, but labelling of classes is introduced**

$$\lambda e_i = S_w^{-1} S_b e_i$$

- **Where:**

$$S_w = \sum_{j=1}^{c} \sum_{i=1}^{n_j} (x_i^j - m_j) \cdot (x_i^j - m_j)^t$$

$$S_b = \sum_{j=1}^{c} (m_j - m) \cdot (m_j - m)^t$$

Pros and cons

- **It takes into account the class labels, thus it is also useful to act as a linear discriminant classifier**

- **Small sample size problem**
  - We need at least D-C samples in order to compute the inverted S_W
  - Solutions
    - Regularization
    - Performing PCA before LDA

- **Scalability**

Example – FisherFaces

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fisher.1

fisher.2

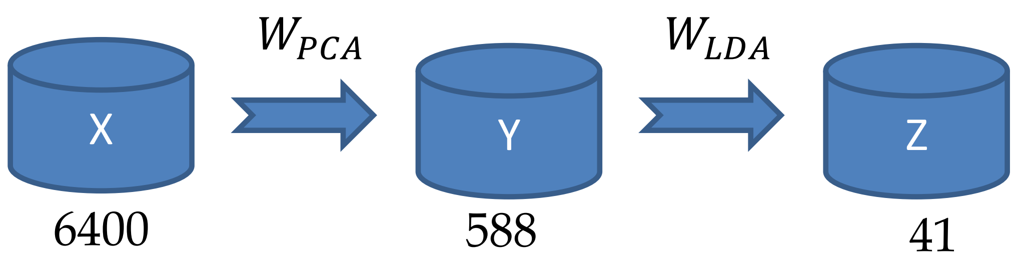
fisher.3

fisher.4

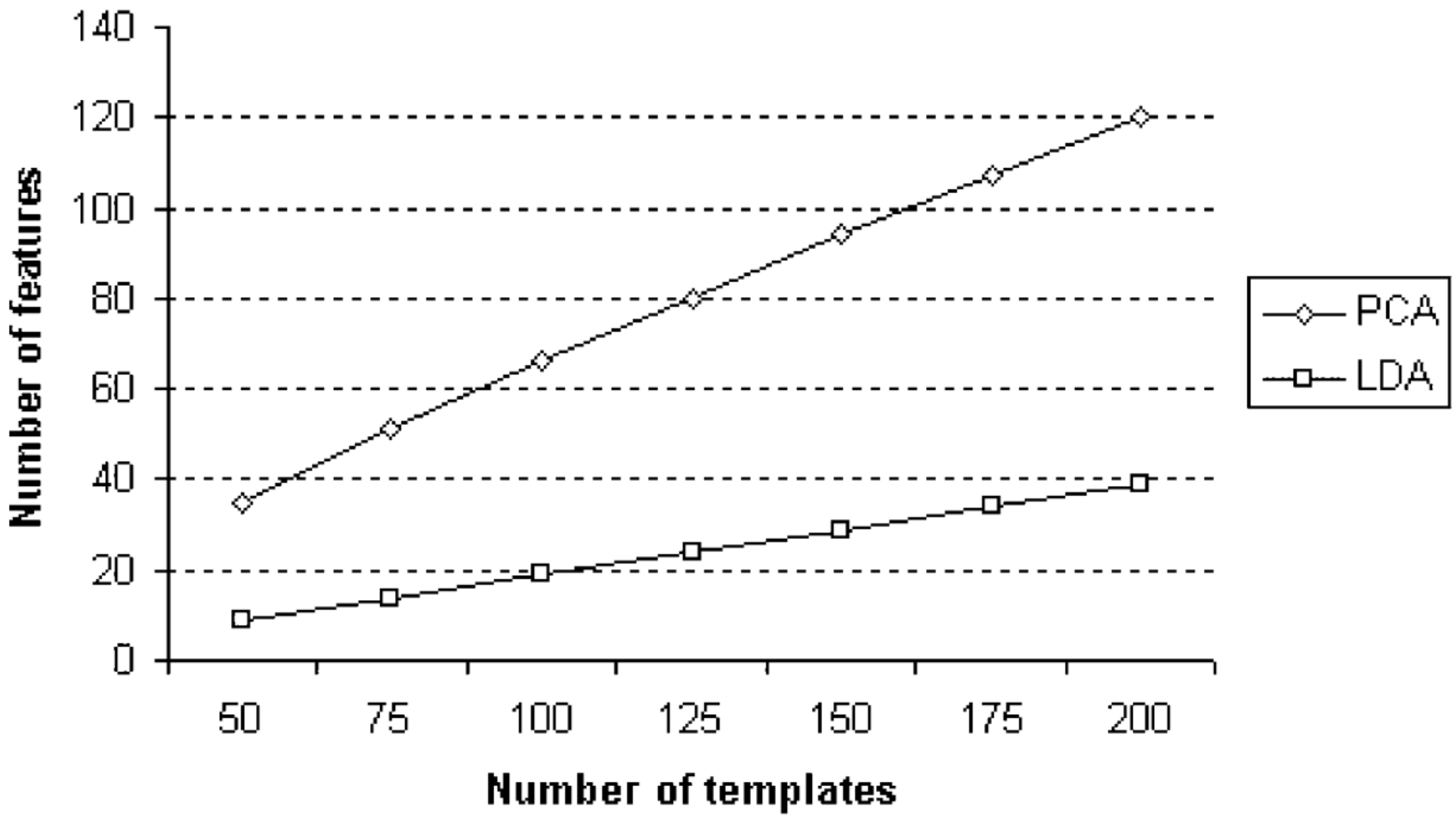
fisher.5

fisher.6

fisher.7



PCA vs LDA : Scalability



Homeworks

- **Kernel-based PCA**
- **User-specific PCA**
- **2D PCA/LDA**
- **ICA**

Homeworks

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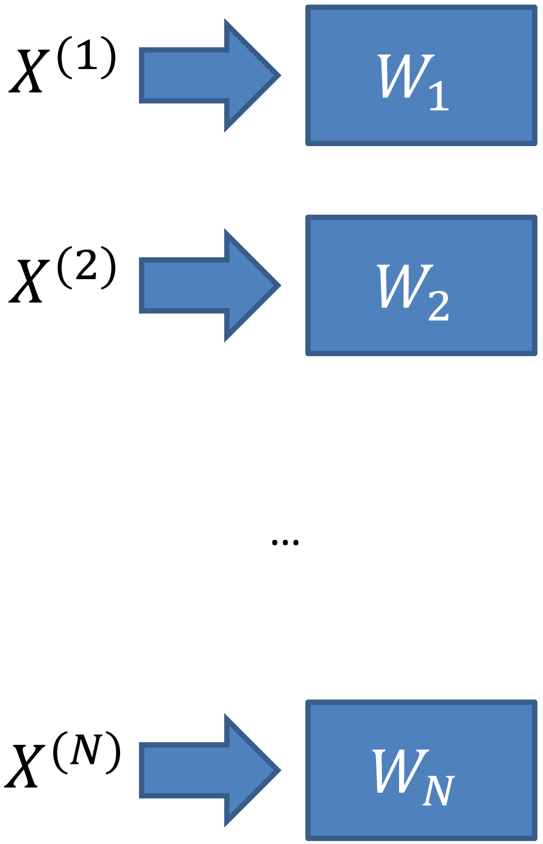
- **Kernel-based PCA**

$$X \xrightarrow[\phi]{} Y$$

- **User-specific PCA**
- **2D PCA/LDA**
- **ICA**

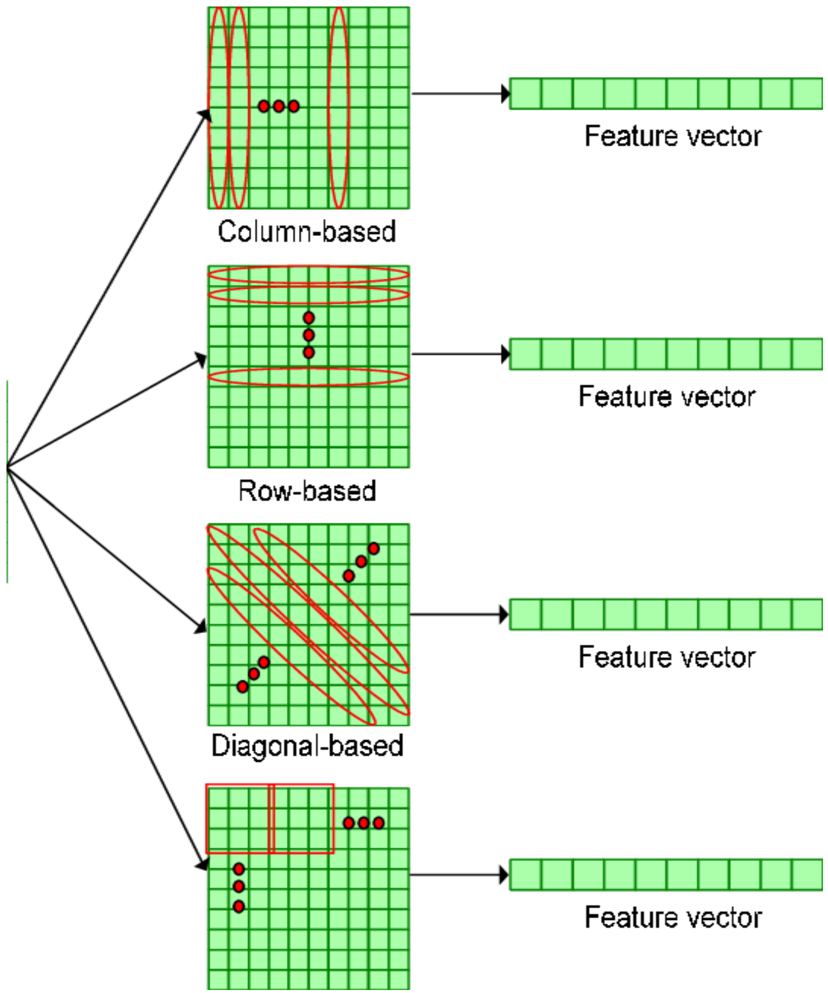
Homeworks

- **Kernel-based PCA**
- **User-specific PCA**
- **2D PCA/LDA**
- **ICA**



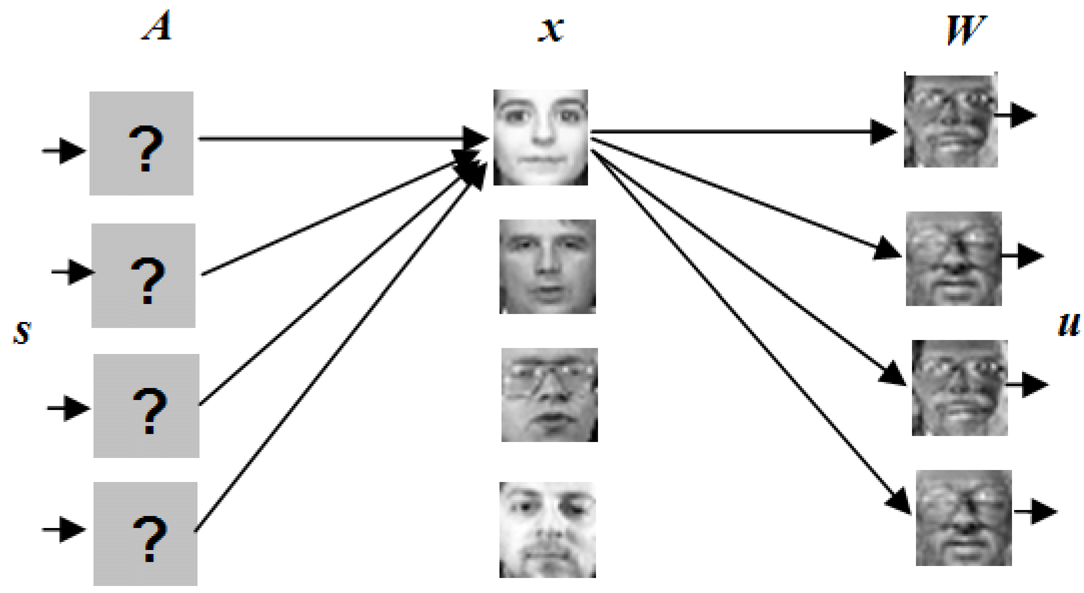
Homeworks

- **Kernel-based PCA**
- **User-specific PCA**
- **2D PCA/LDA**
- **ICA**



Homeworks

- Kernel-based PCA
- User-specific PCA
- 2D PCA/LDA
- ICA



Homework

- **Given a set of faces--- we will make available a data set in the Teams platform**

- **Compute PCA over the set of faces**

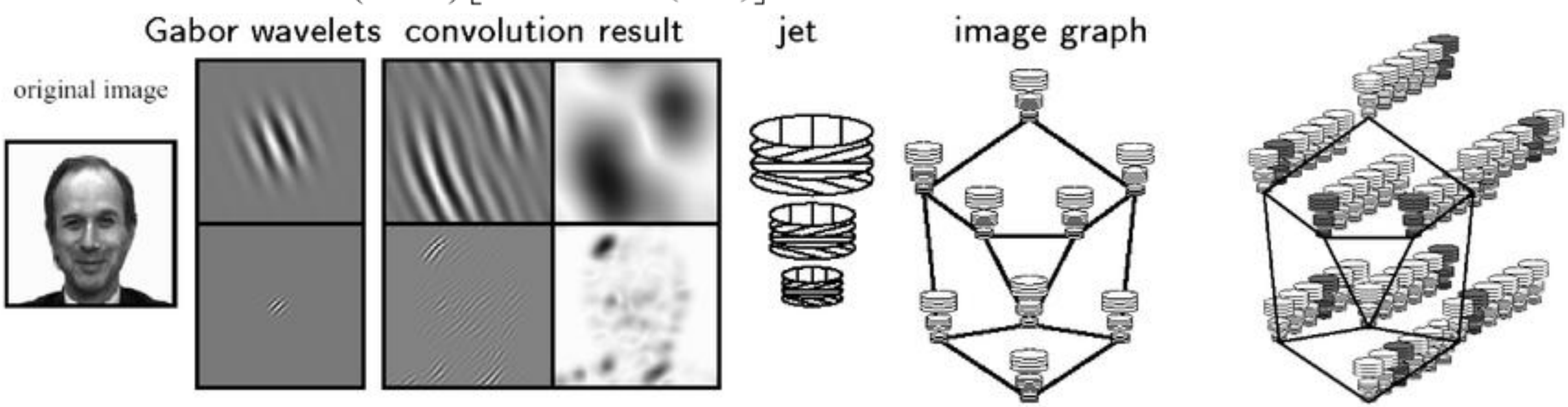
- **Plot the average distances among images of the same subject according to increasing values of p**

- **Plot the average distances among images of different subjects according to increasing values of p**

- **Check the best trade-off**

Elastic Bunch Graphs

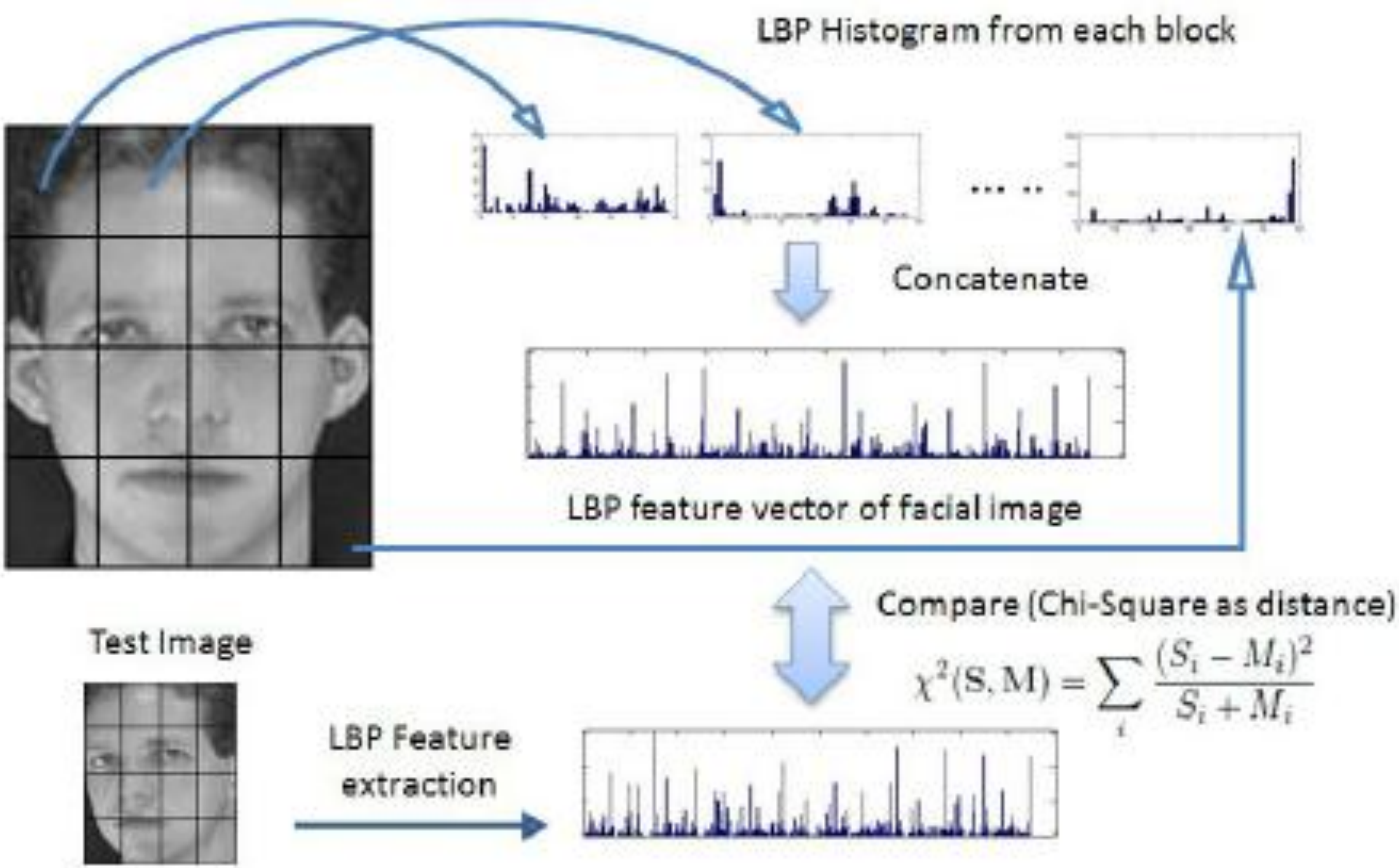
$$\psi_j(\vec{x}) = \frac{k_j^2}{\sigma^2}\exp\left(-\frac{k_j^2 x^2}{2\sigma^2}\right)\left[\exp\left(i\vec{k}_j\vec{x}\right) - \exp\left(-\frac{\sigma^2}{2}\right)\right]$$



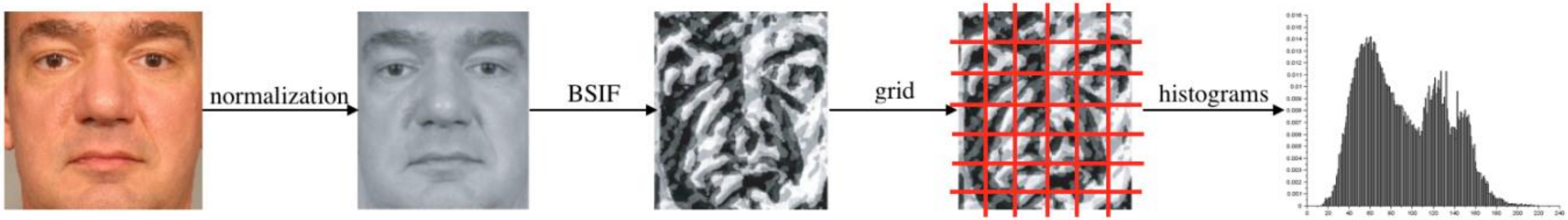
(a) Elastic Graph Representation

(b) Bunch Graph

Local Binary Patterns



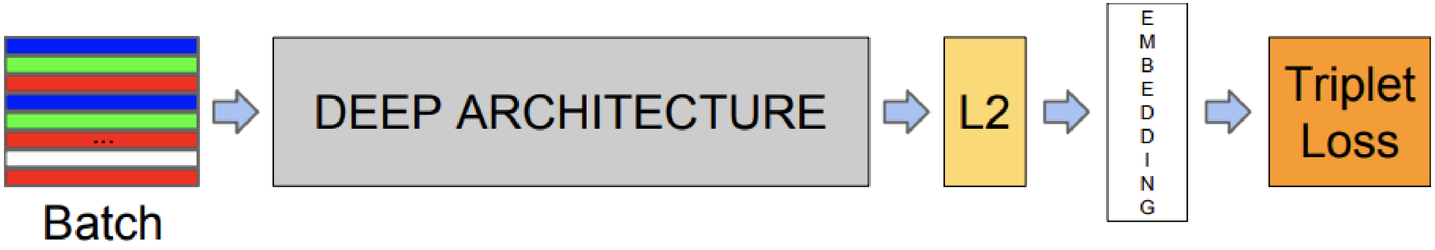
Generalization to textural features

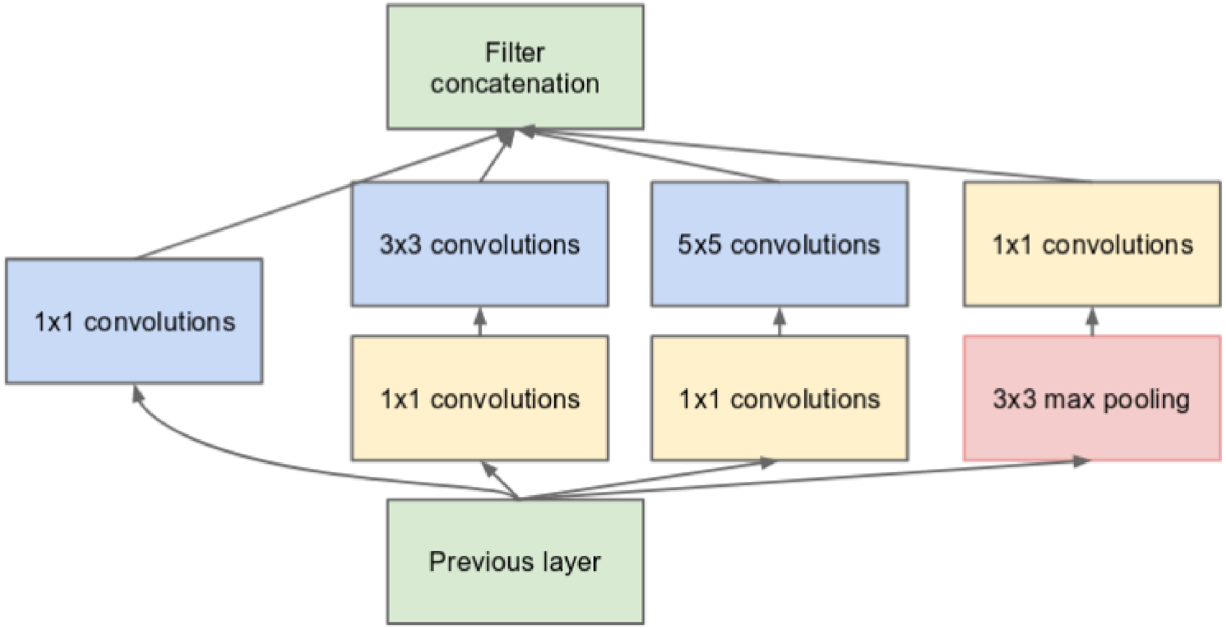


Be careful!

Be careful!

Deep learning-based approaches: FaceNet





Triplet loss function

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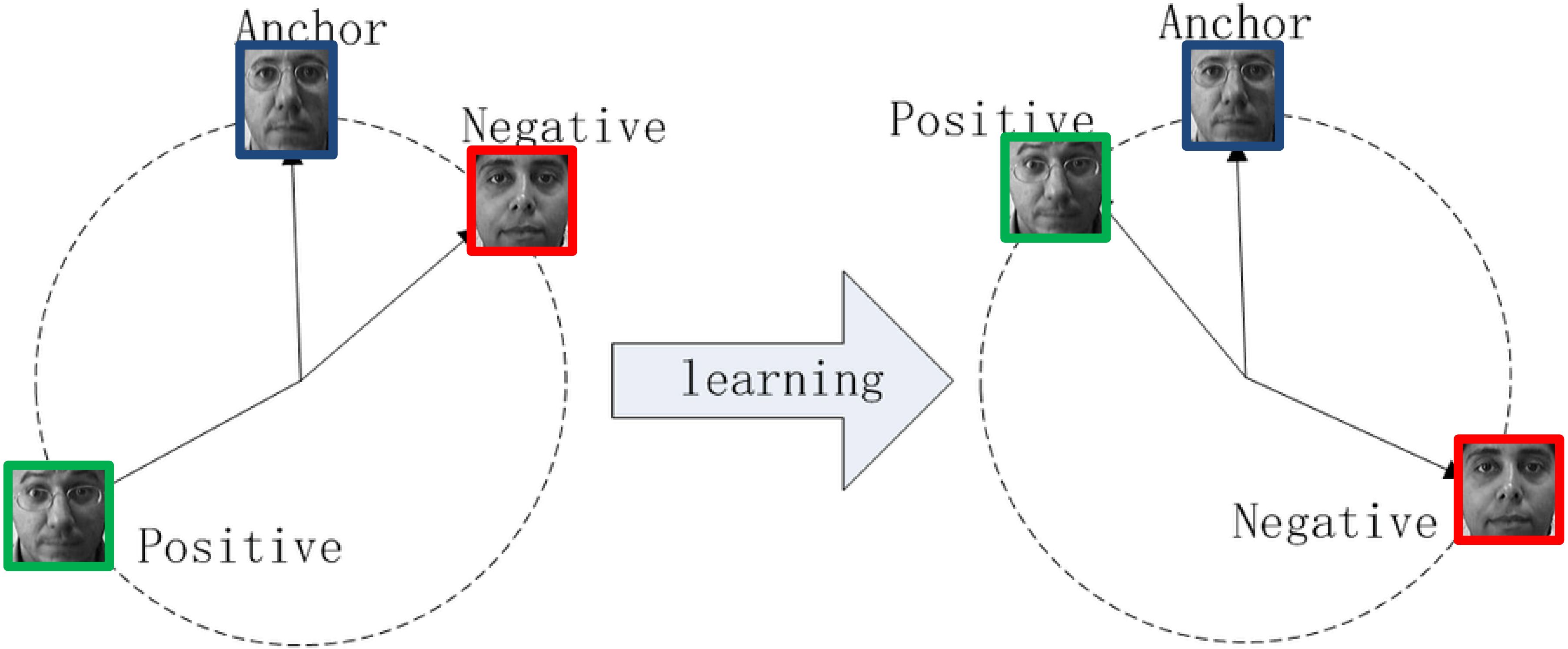

- Distance between the Anchor and the Positive is **minimised**
- Distance between the Anchor and the Negative is **maximised**

Homework

- Download the software for LBP, LPQ, BSIF, HOG, SIFT methods
- Obtain the correspondent LBP/LPQ/BSIF… images
- Design a tassellation ==> can you suggest a better one than that shown here?
- Train and test a Pairwise-SVM
- Download the pre-trained models of FaceNet
- Test them on a publicly available facial data set

Face matching

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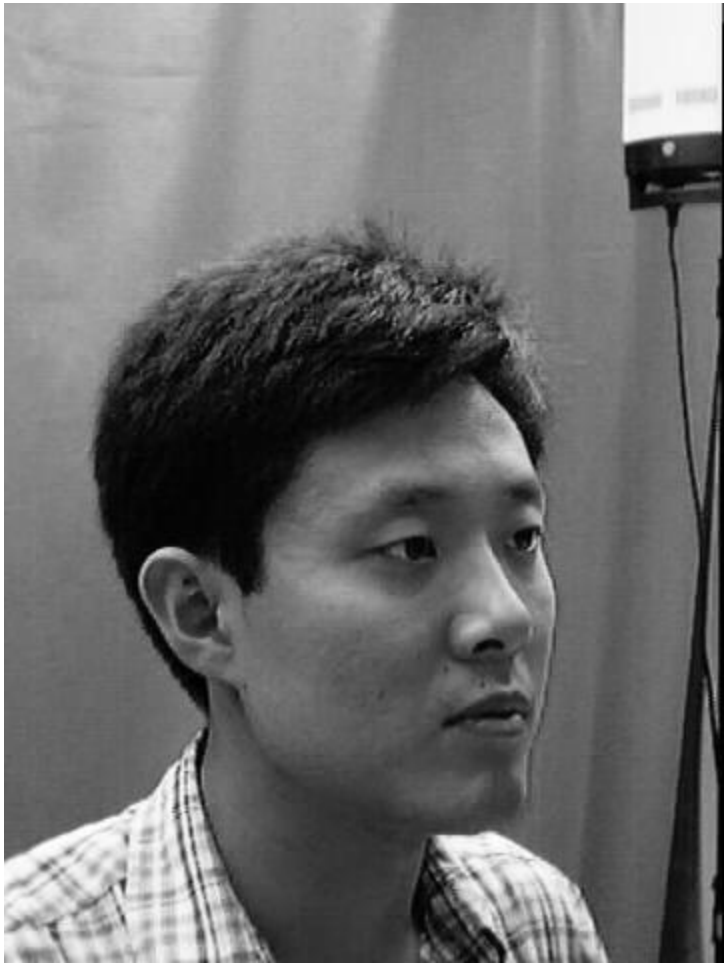
- **Distances according to the feature space**
  - Hamming
  - Hellinger
  - Euclidean
  - Mahalanobis…

- **Graphs (EBGM, Active Shape Models)**
  - Appropriate distance are defined

Elastic bunch graph distance



$$S_v\left(J^I, J^M\right) := \frac{\left|J^I\right| \cdot \left|J^M\right|}{\left\|J^I\right\| \left\|J^M\right\|},$$

$$S_e\left(\vec{\Delta}_{ij}^{I}, \vec{\Delta}_{ij}^{M}\right) := -\frac{\left\|\vec{\Delta}_{ij}^{I} - \vec{\Delta}_{ij}^{M}\right\|^2}{\left\|\vec{\Delta}_{ij}^{M}\right\|^2},$$

$$S_G(G^I, G^M) := \sum_{i \in V} S_v(J_i^I, J_i^M) + \lambda \sum_{(i,j) \in E} S_e\left(\vec{\Delta}_{ij}^{I}, \vec{\Delta}_{ij}^{M}\right).$$

Distances in the feature space

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- **Appearance-based methods**
  - Residual face module
  - Transformed image vs. original image
  - Euclidean distance in the novel feature space

- **Textural-based methods**
  - L1 distance
  - Histogram vs. histogram

$$d(H_1, H_2) = \sum_{i=0}^{2^n-1} |H_1(i) - H_2(i)|$$

Score transformation function

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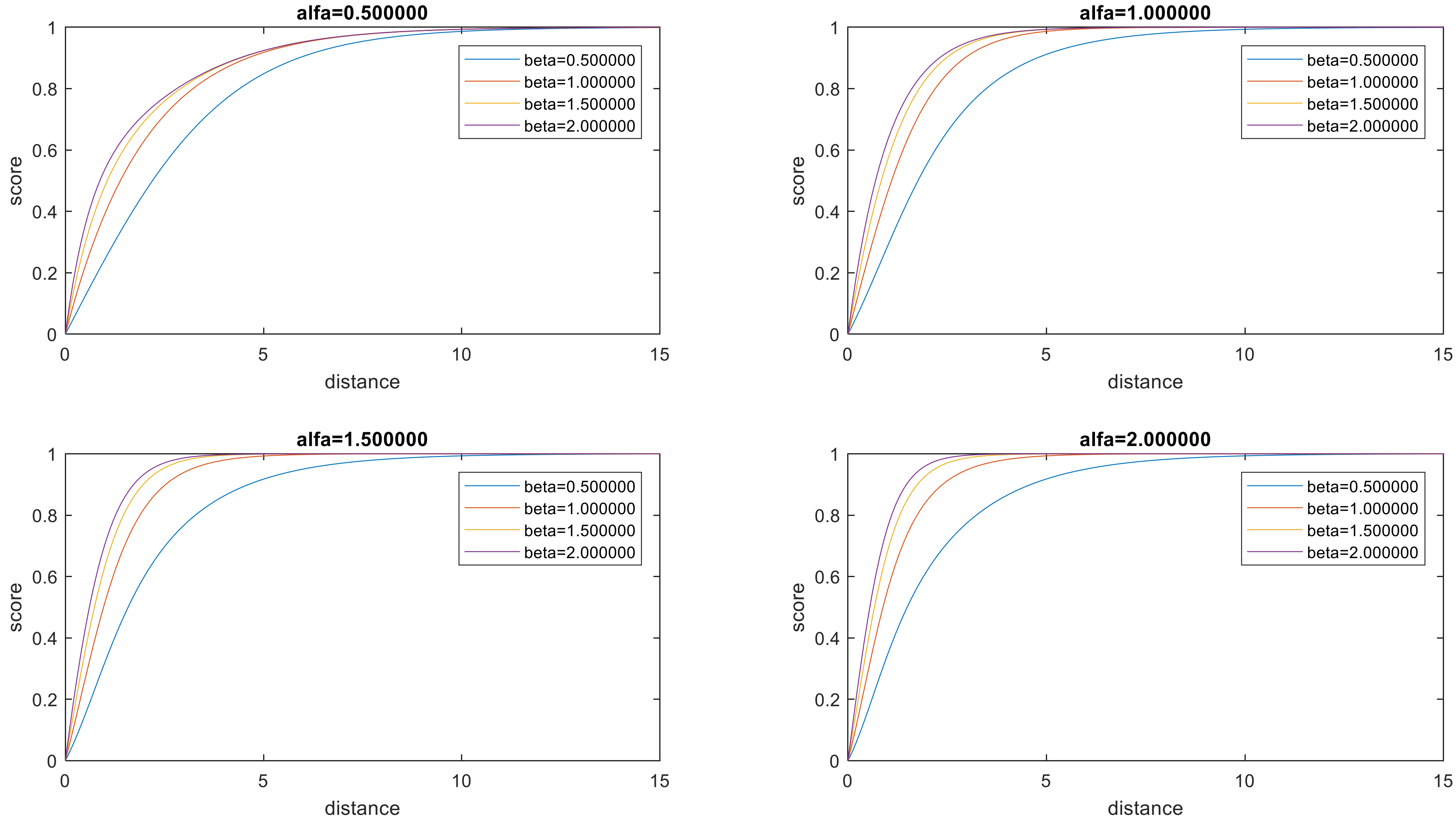
- **Linear normalization (stretch)**
  - Validation set needed
  - $s = \frac{maxd - d}{maxd - mind}$
- **Non-linear normalization**
  - $s = \frac{1}{d+1}$; Z-score (validation); Mahalanobis
- **Pseudo-Logistic normalization**
  - $s = \frac{1-\exp(-\beta \cdot d)}{1+\exp(-\alpha \cdot d)}$

Pseudo-logistic normalization



Experimental comparison

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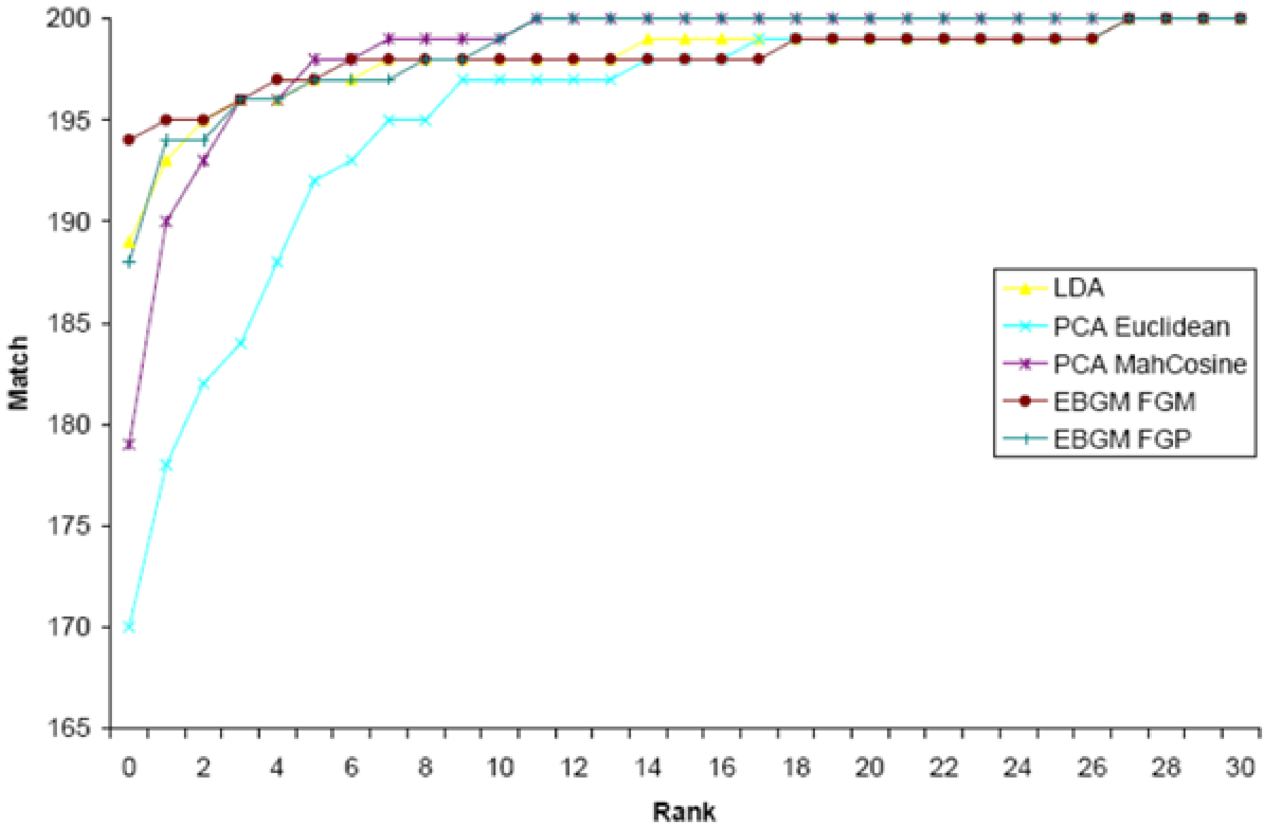


Further info:

Oloyede, M.O., Hancke, G.P. & Myburgh, H.C.: A review on face recognition systems: recent approaches and challenges. *Multimedia Tools and Applications* 79, 27891–27922 (2020). https://link.springer.com/article/10.1007/s11042-020-09261-2

Homework

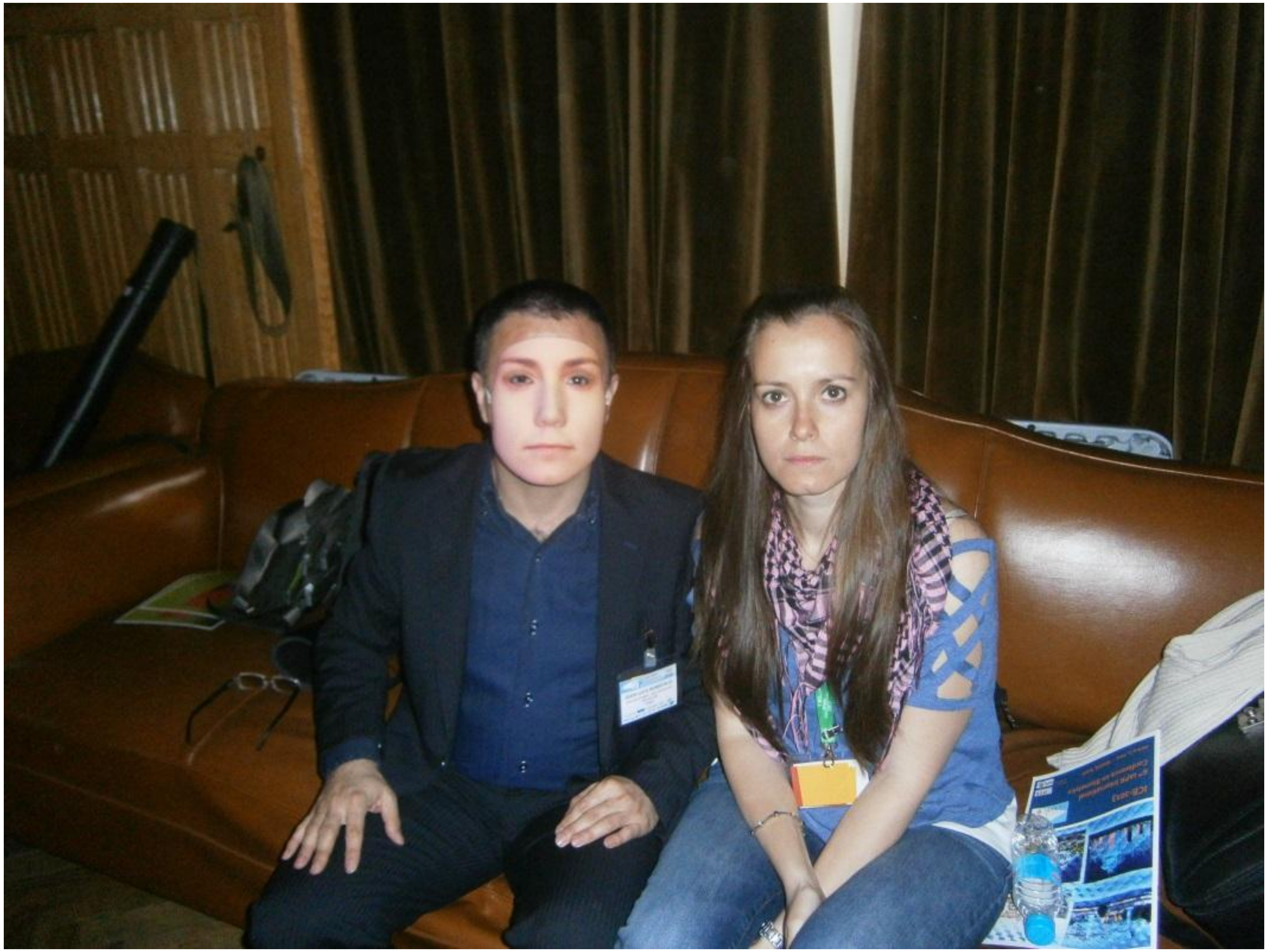
- **Download or implement source-code for feature extraction from facial images**

- **Holistic- or local-based approach?**
  - LBP on the whole face?
  - LBP computed separately on a tassellation?

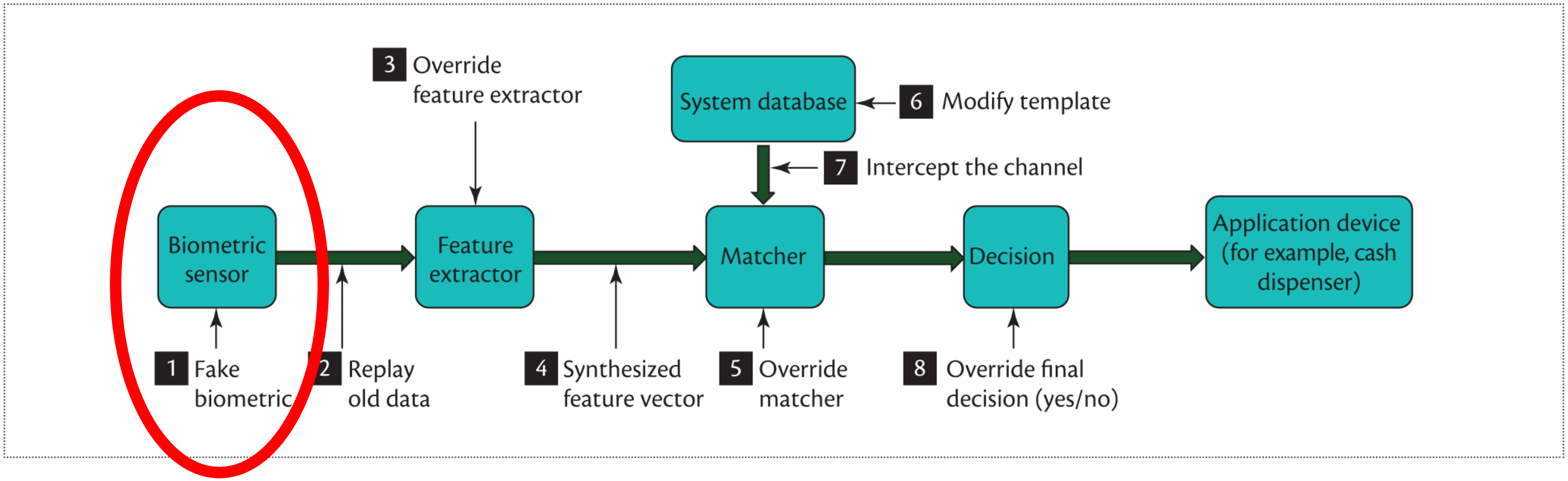
- **Using a publicly available face data set, explore the difference**

Issues in face recognition
Presentation attacks detection

Presentation attacks detection



Vulnerability points



Possible attacks

Picture

Video

Mask

Goals

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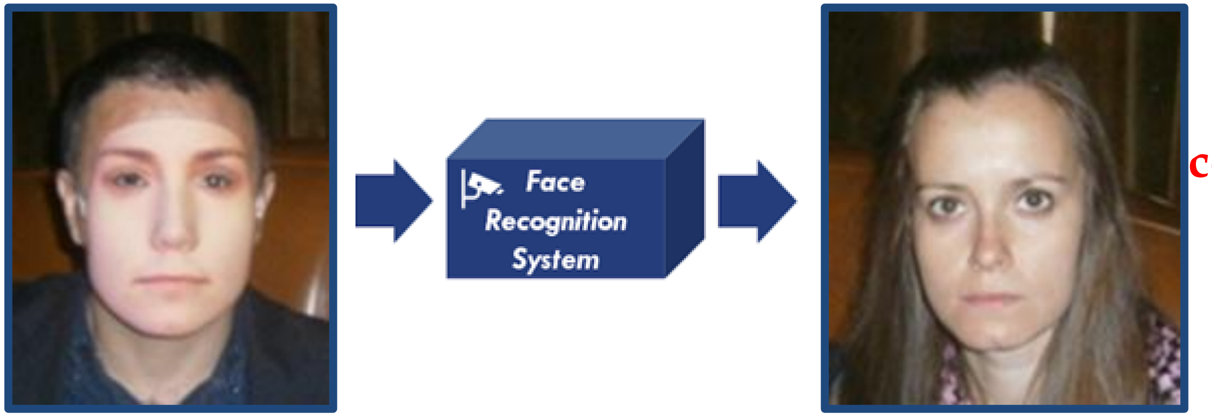
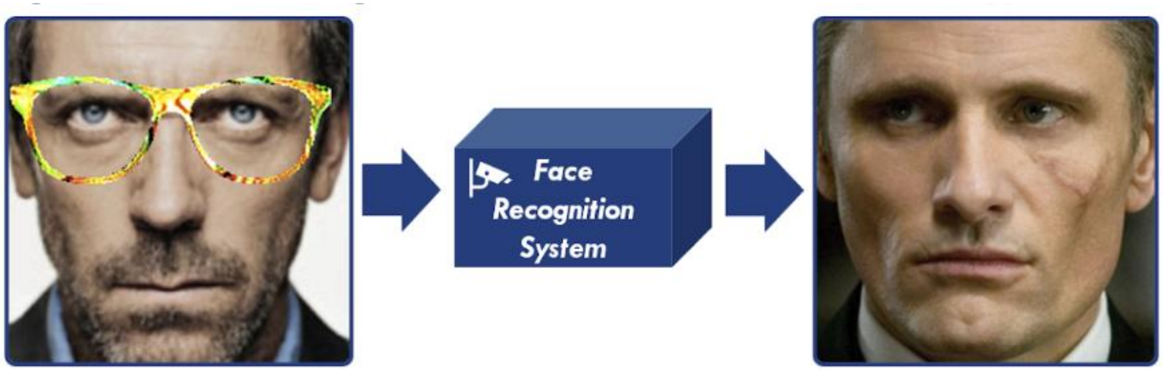
- «Empowered» impostors: being misclassified as a genuine user



Identity confirmed
Access granted

- Evade («obfuscate») recognition: being classified as someone else



The subject is not in the «black list»

Presentation Attack Detection

Training samples of live and fake face images/videos

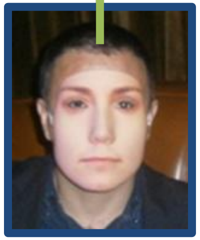
SENSOR → **FEATURE EXTRACTOR** → **CLASSIFIER**

Liveness score

Threshold

LIVE

FAKE

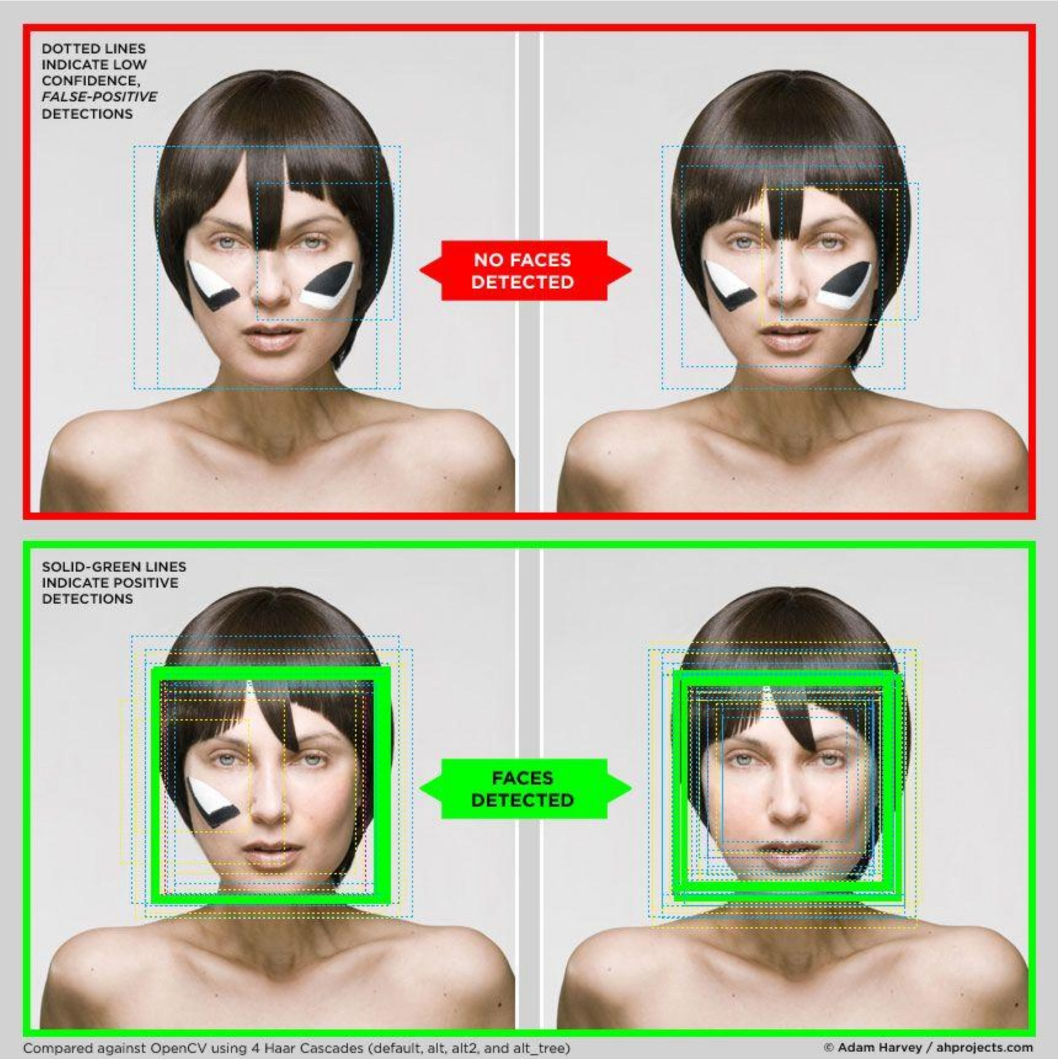


Presentation attack

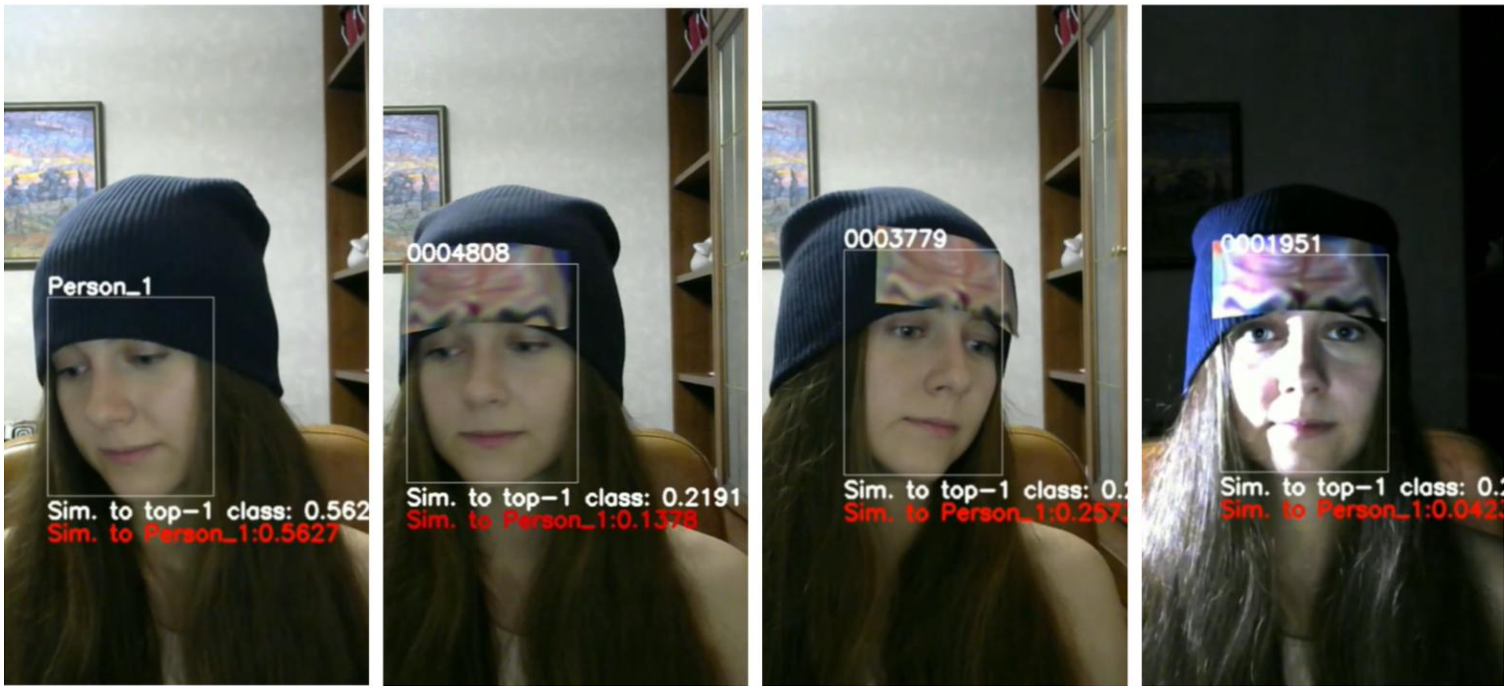
Presentation attacks detection approaches

- **Skin properties**
  - Texture, reflectance
  - LBP, BSIF, Wavelets
- **Depth + texture**
  - By 3d cameras (Kinect)
- **Eye blinking and mouth movements**
  - Video-based only
- **Artifact detection**
  - Due to the fake fabrication procedure

Evasion based on avoiding face detection



Camuflage and make-up



In this and previous cases, the problem is to detect the artifact: what is out of context ?

Homework

- **Can be appearance-based methods of some helps in detecting presentation attacks from smartphones?**
- **In your opinion, how should PCA be trained?**
- **Collect your personal live-fake data set using free vips images and then re-acquiring them by using both smartphone and webcam**
- **Design a fake detector based on principal component analysis**
  - Select the training set appropriately
- **Train and test by**
  - Template based match
  - Support Vector Machine

«Second-level» presentation attacks

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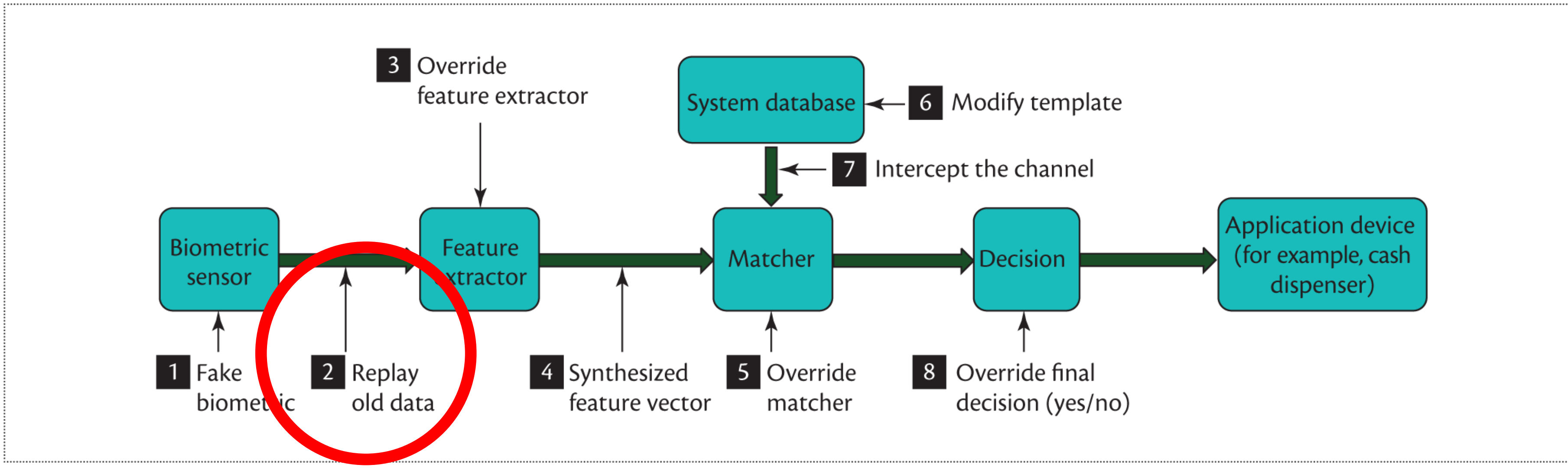
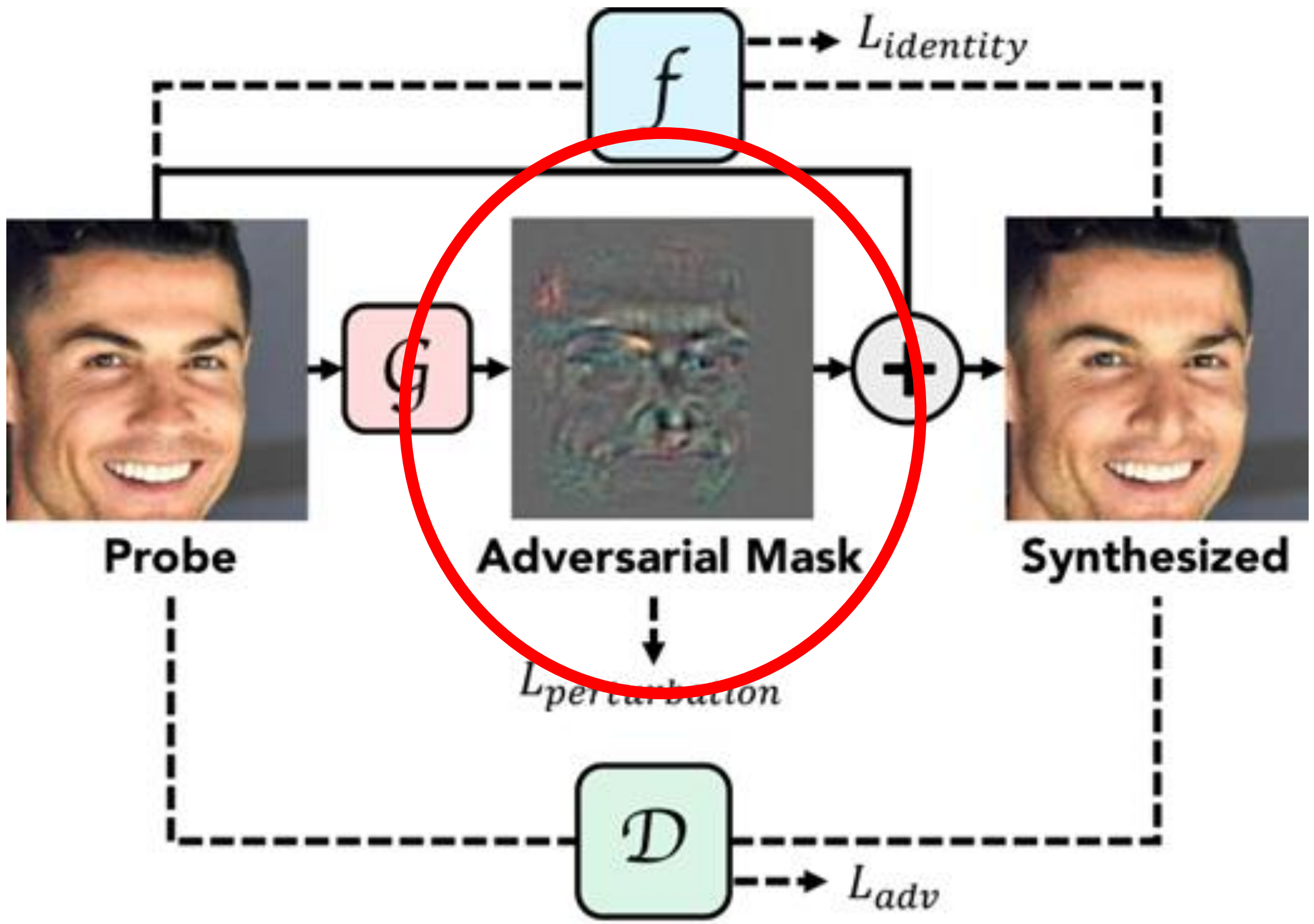


«Second-level» presentation attacks: adversarial deep learning for evasion



Deep fakes

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Replicating the facial appearance of a certain individual in order to impersonate someone else in a video-sequence

Generative Adversarial Networks

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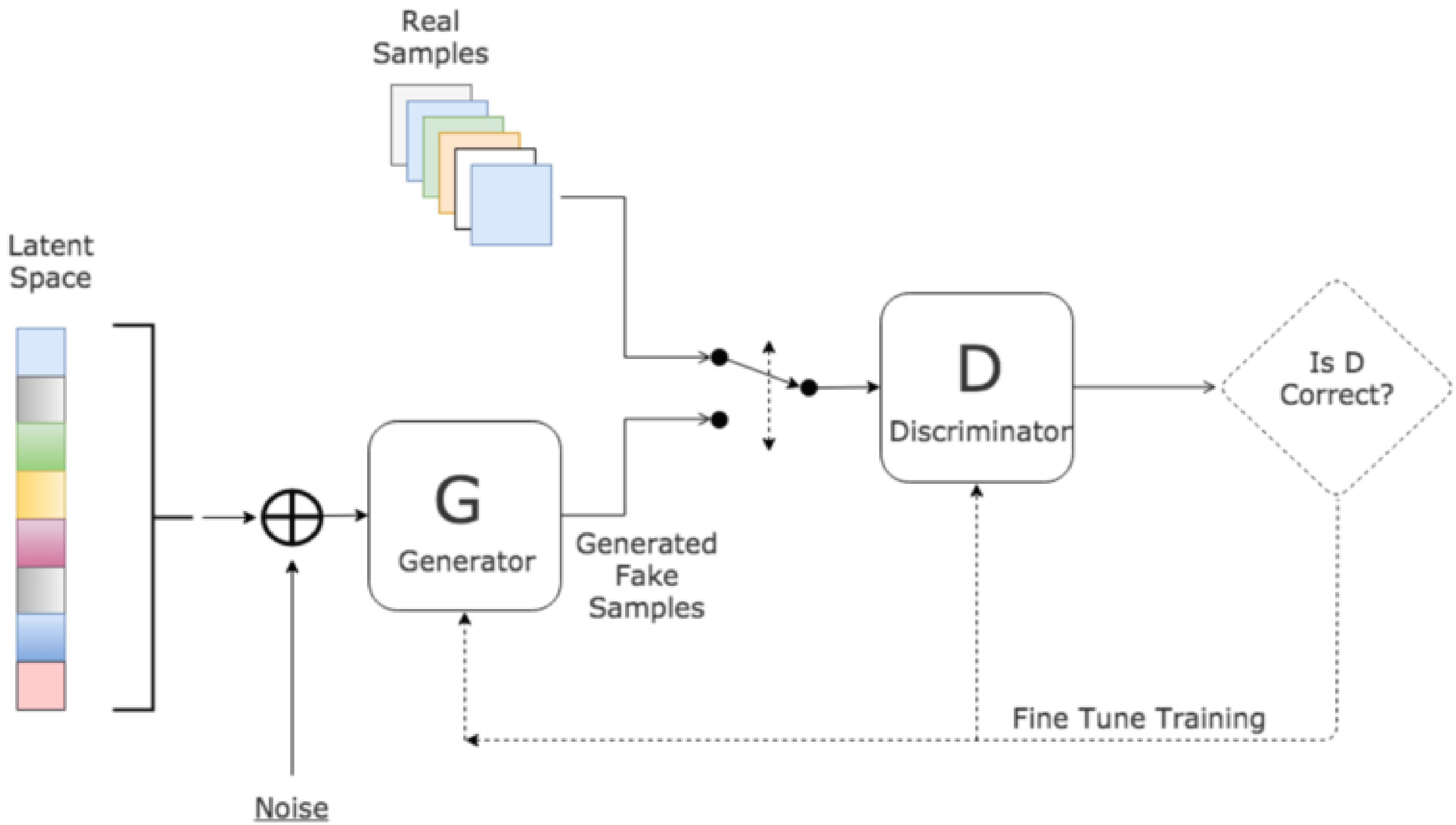

Deep fakes by facial swap

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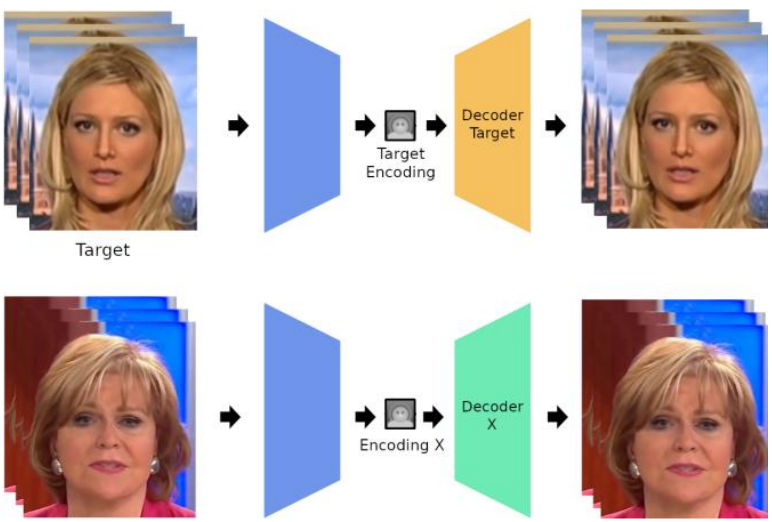
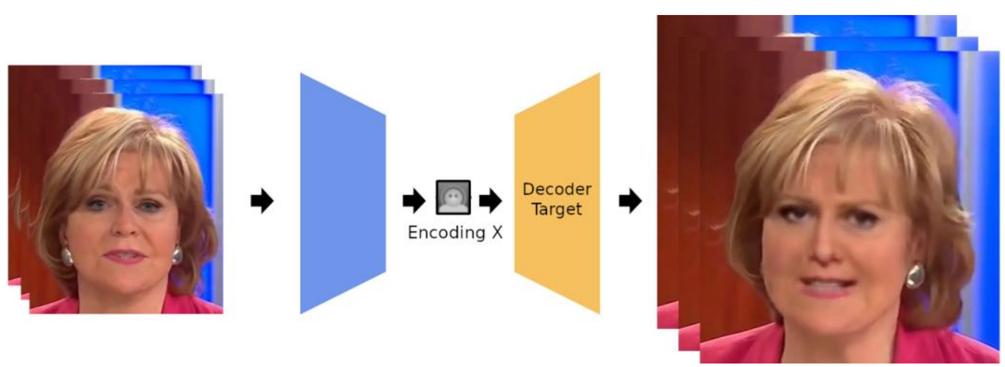
Transferring and manipulating the target face on the one of a novel subject



Training step

Test step



Deep fakes by facial reenactment

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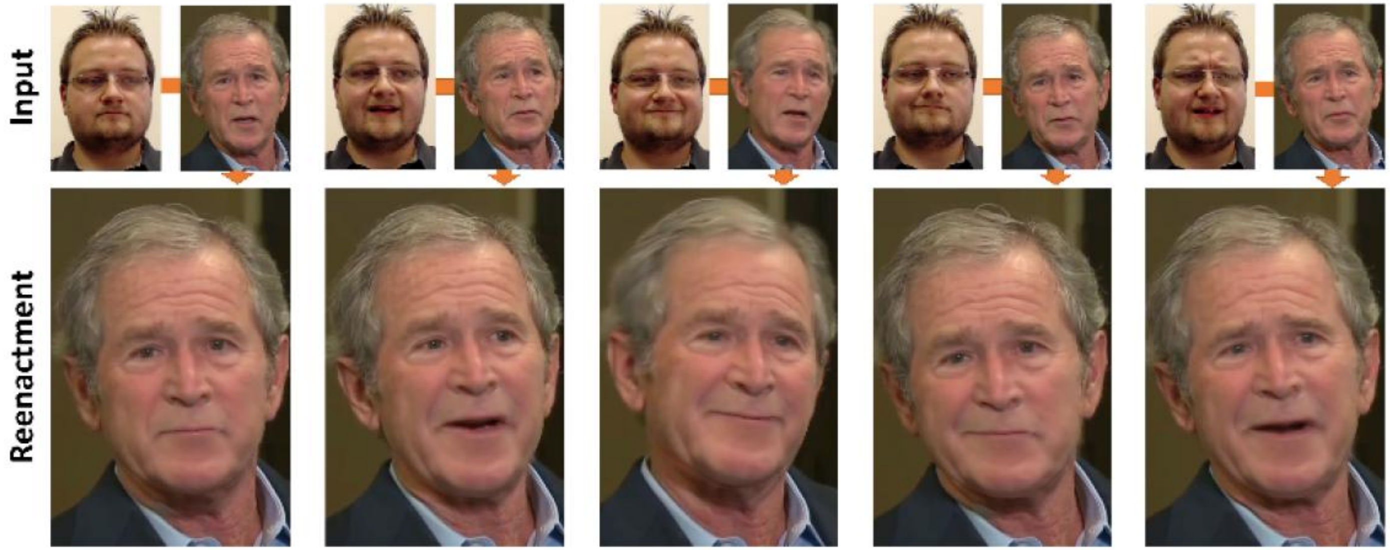
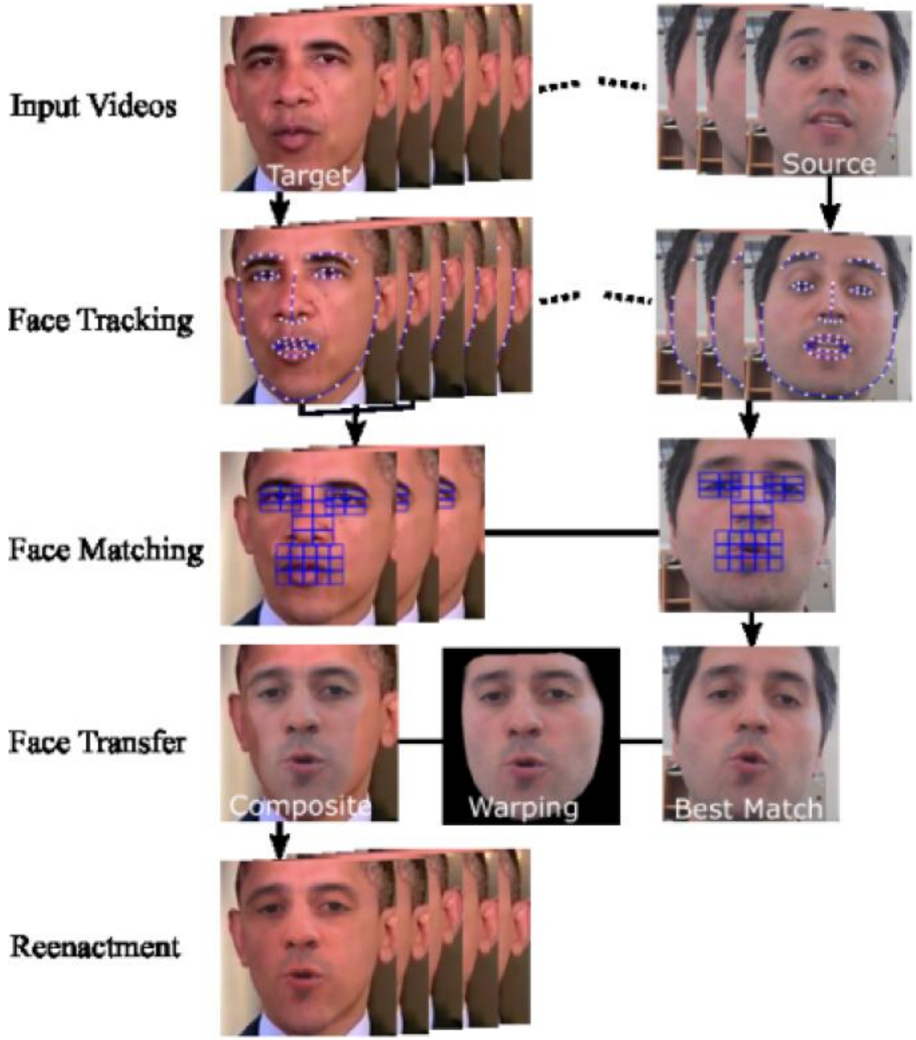
Manipulation of the target face in her/his own domain



Facial reenactment: steps



Correspondent landmarks

High-level features

Deep fake detection challenge

	UADFV [39]	DF-TIMIT [23]	FF++ [31]	DFD [30]	DFDC [8]	Celeb-DF [40]
Two-stream [41]	85.1	73.5	70.1	52.8	61.4	53.8
Meso4 [6]	84.3	68.4	84.7	76.0	75.0	54.8
MesoInception4	82.1	62.7	83.0	75.9	73.2	53.6
HeadPose [39]	89.0	53.2	47.3	56.1	55.9	54.6
FWA [26]	97.4	93.2	80.1	74.3	72.7	56.9
VA-MPL [27]	70.2	62.1	66.4	69.1	61.9	55.0
VA-LogReg	54.0	77.3	78.0	77.2	66.2	55.1
Xception-raw [31]	80.4	54.0	**99.7**	53.9	49.9	48.2
Xception-c23	91.2	94.4	99.7	**85.9**	72.2	65.3
Xception-c40	83.6	70.5	95.5	65.8	96.7	**65.5**
Multi-task [29]	65.8	55.3	76.3	54.1	53.6	54.3
Capsule [28]	61.3	74.4	96.6	64.0	53.3	57.5
DSP-FWA	**97.7**	**99.7**	93.0	81.1	**75.5**	64.6

Issues in face recognition
Template Updating

Template Updating

INTER-CLASS SIMILARITY

INTRA-CLASS VARIATIONS

VARIATIONS OVER TIME

Partial solution:

- **Multiple supervised re-enrollment**

expensive

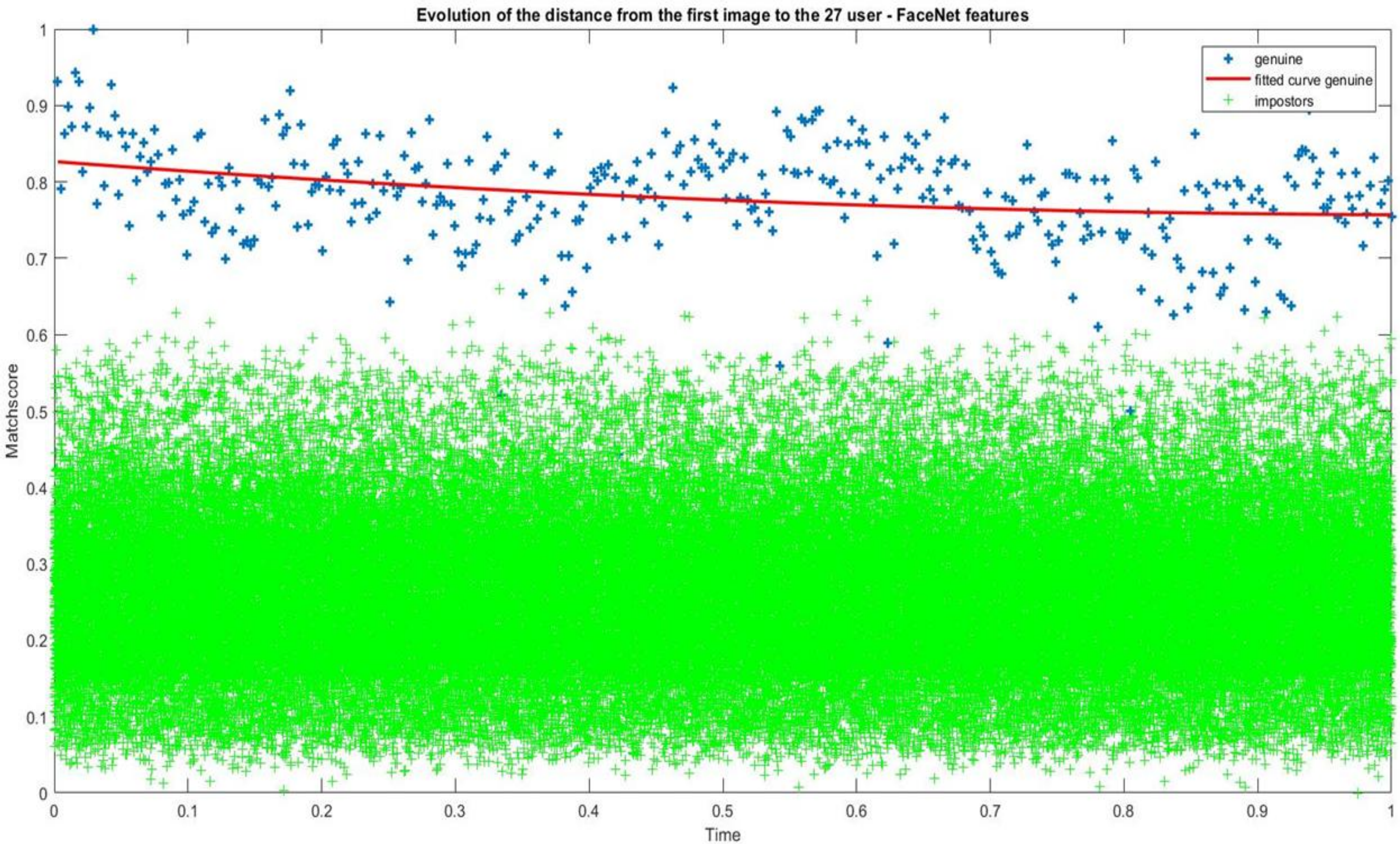
time-consuming

invasive

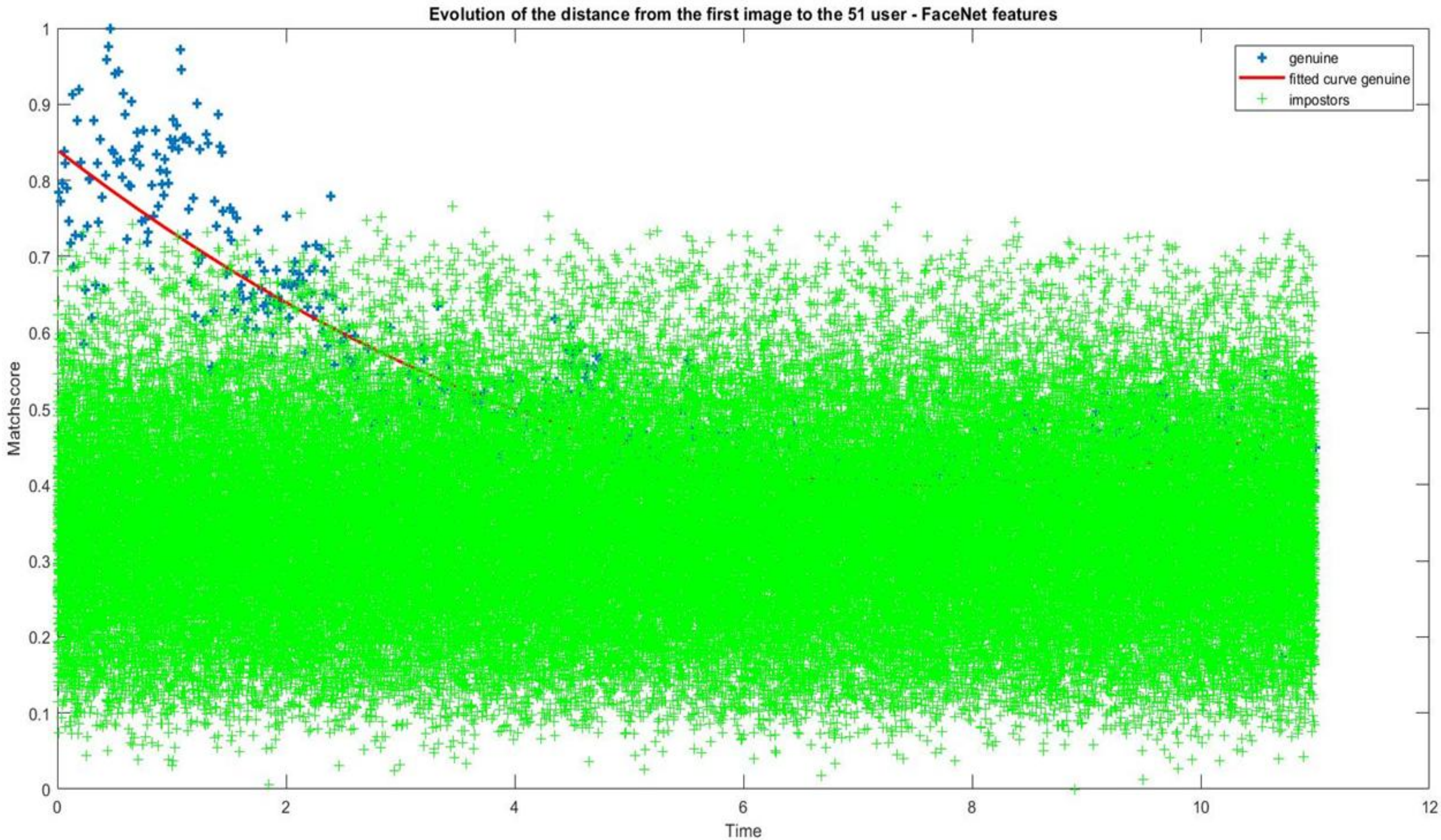
TEMPLATE

TEST IMAGES

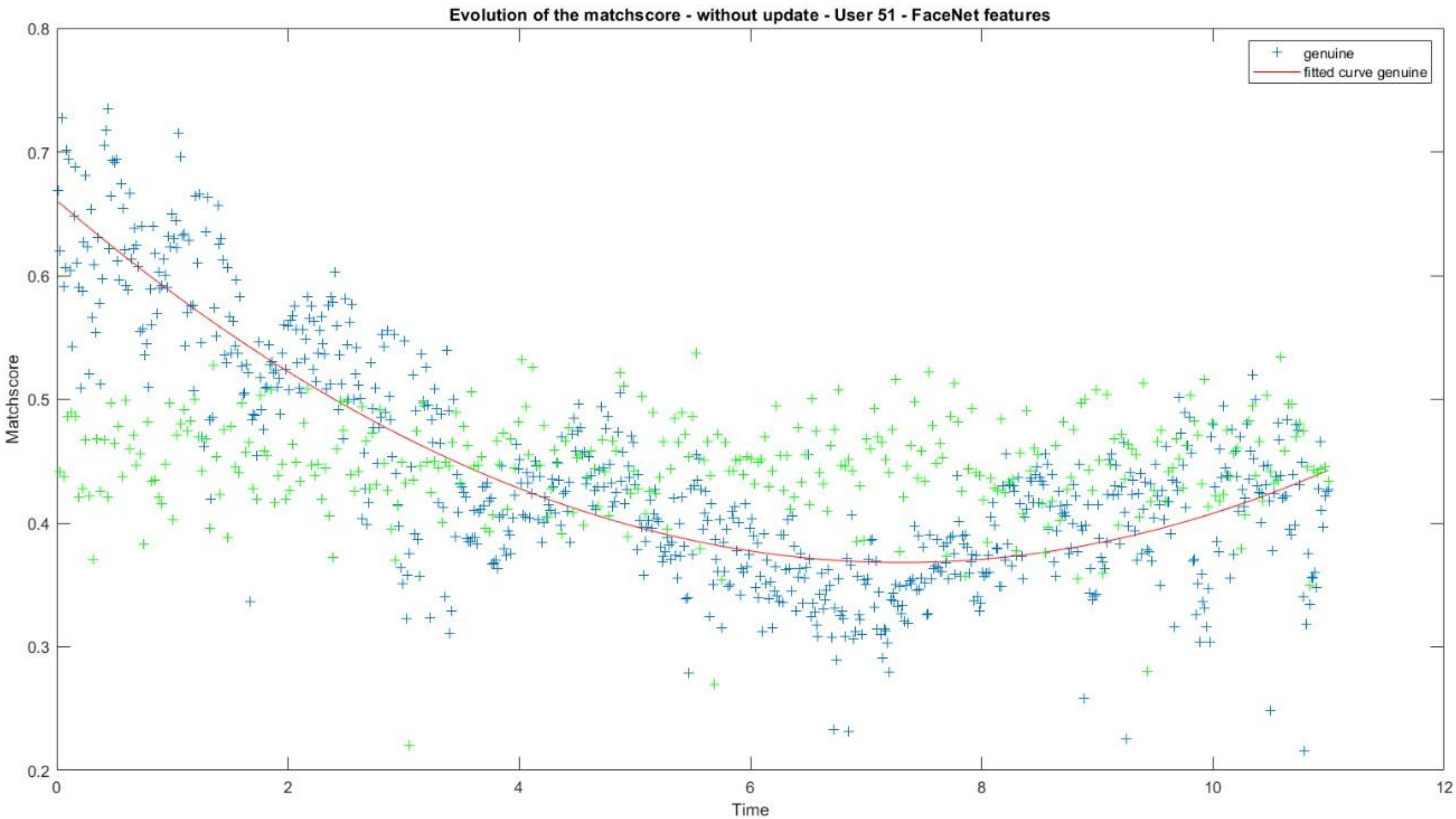
Motivations: «strong» people recognition



Motivations: «weak» people recognition



A deeper view



Adaptive systems

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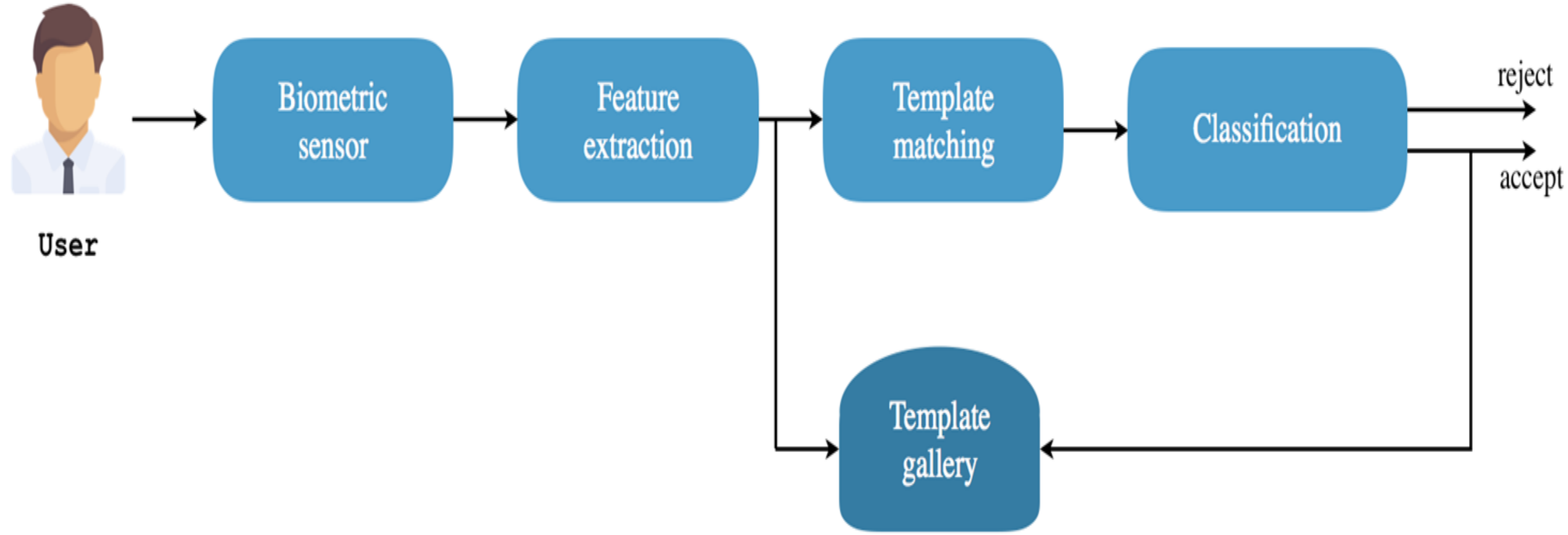
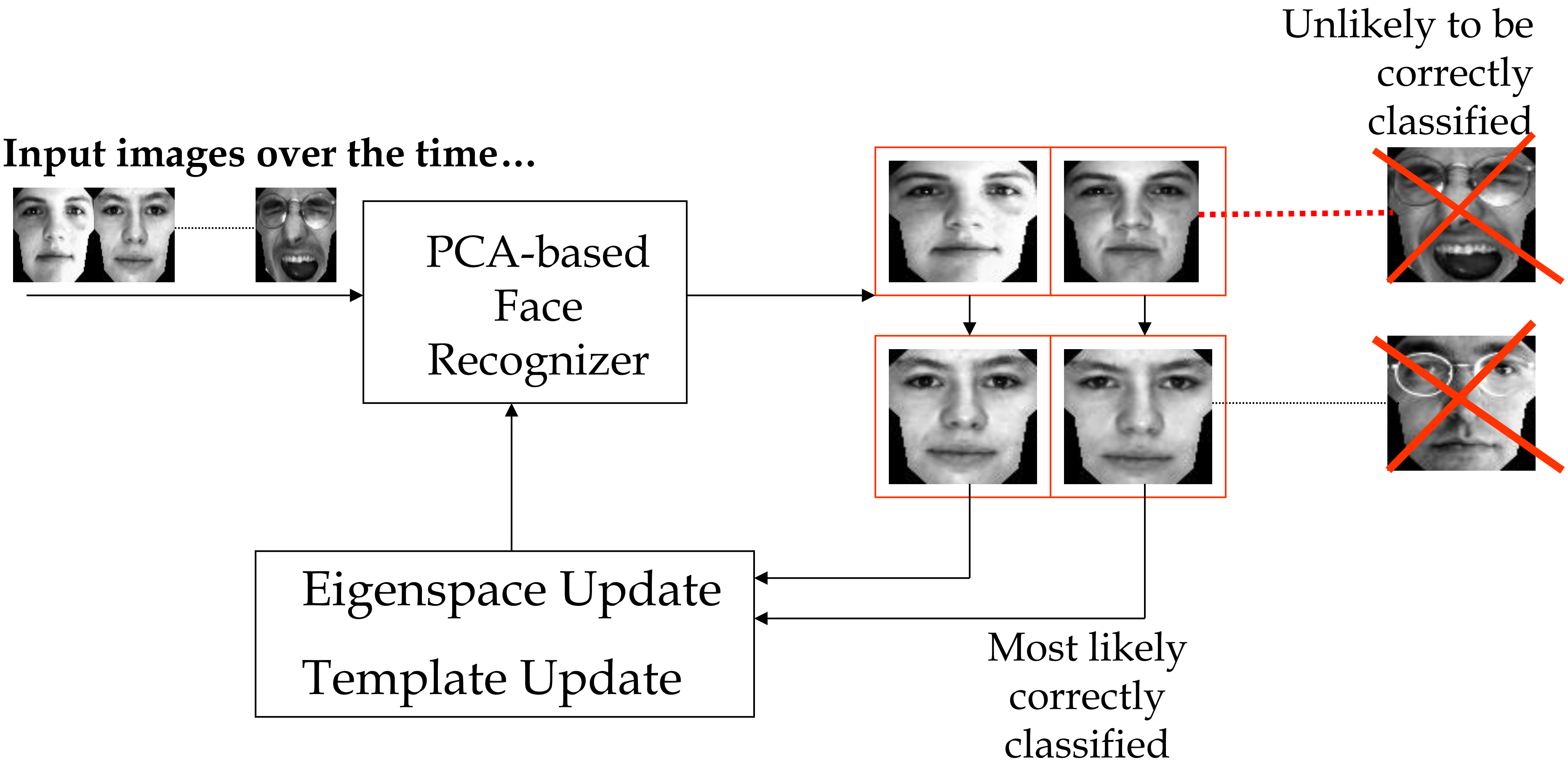
- Systems able to update/add templates without the need of a supervisor
- They exploit data acquired during system's operation



A case-study: PCA-based face recognition



Gallery increase and «impostors» drift

Incremental update of some users' galleries	Initial Accuracy	After Semi-super-vision
	22.9%	94.3%
	14.3%	49.0%
	32.7%	95.9%
	14.3%	85.7%

Approaches to template update

CLASSIFICATION
Incremental introduction of novel samples into the «galleries»

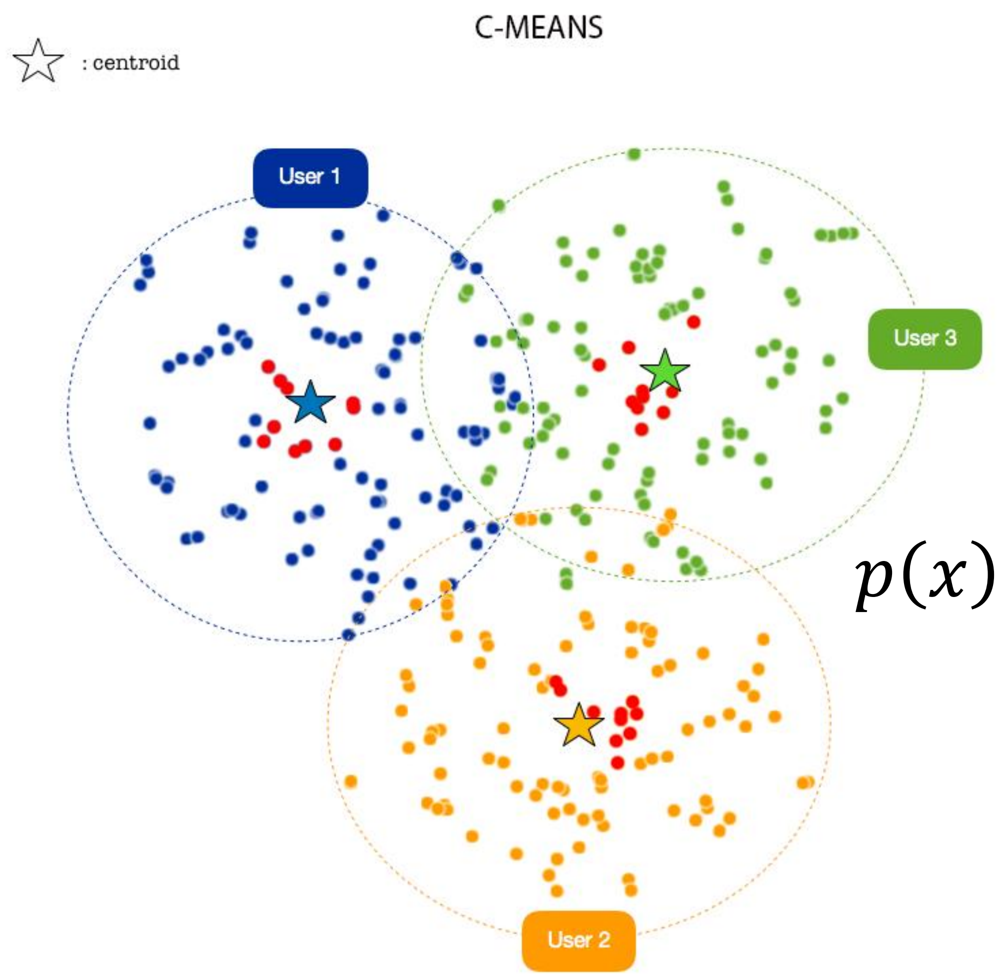
CLASSIFICATION-SELECTION
Incremental introduction and **pruning**

Generic user gallery after classification

Template update algorithms

- **Classification with unlimited gallery size (ideal)**
  - Appearance-based with incremental training (incremental PCA)
  - Machine learning-based (incremental SVM)
- **Classification-selection with unlimited gallery size (unlikely but better)**
  - Risk-based optmization
  - Context sensitive
- **Classification-selection with limited gallery size (real)**
  - FIFO, LRU, LFU
  - MDIST, DEND
  - Clustering-based

Classification-selection by centroid detection

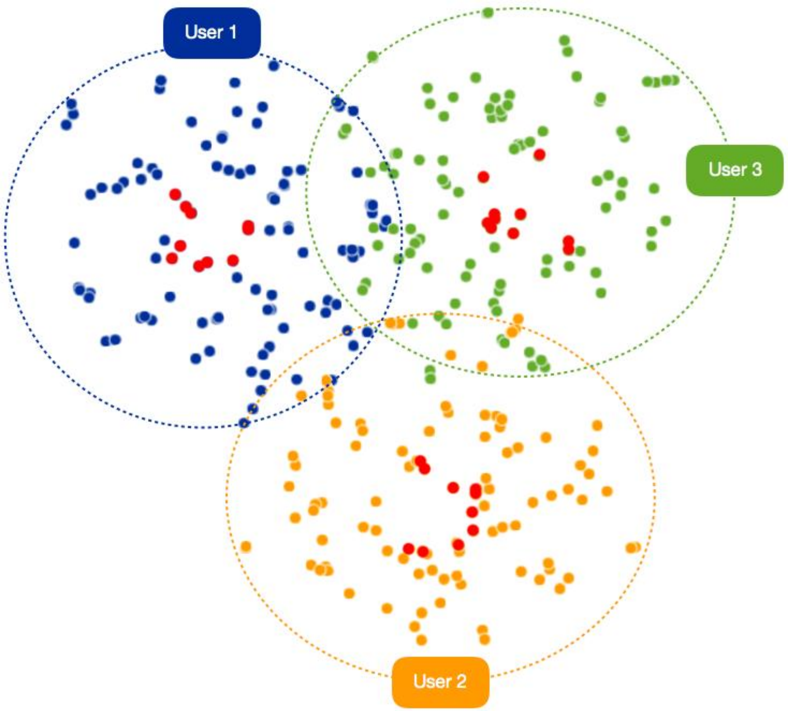


$$p(x) = \sum_{i=1}^{n} p(x|user_i) \cdot P(user_i)$$

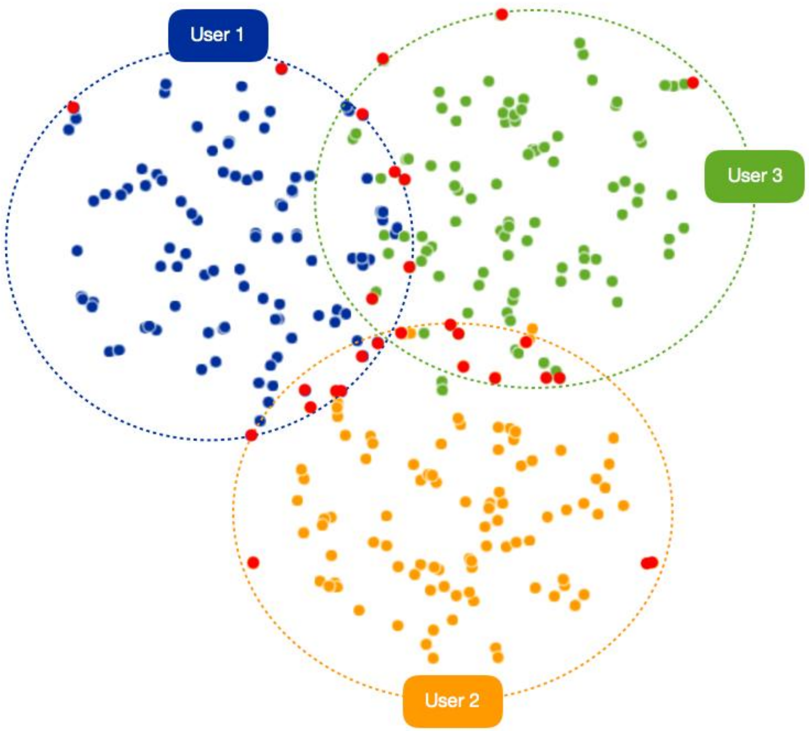
$$p(x|user_i) \sim N(\mu_i, \Sigma_i)$$

Classification-selection by intra-class variations

Minimization
(MDIST)

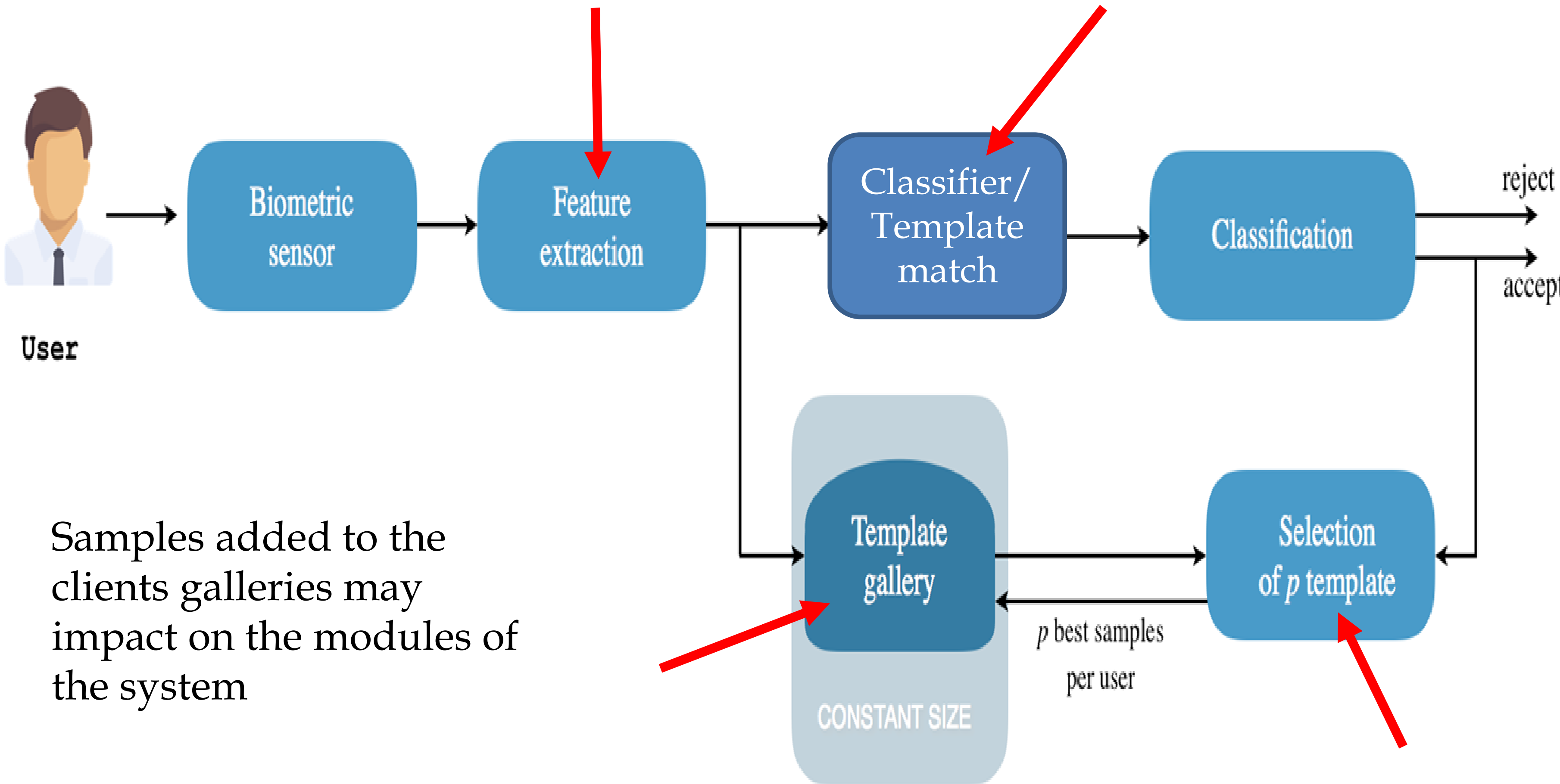


Maximization
(DEND)



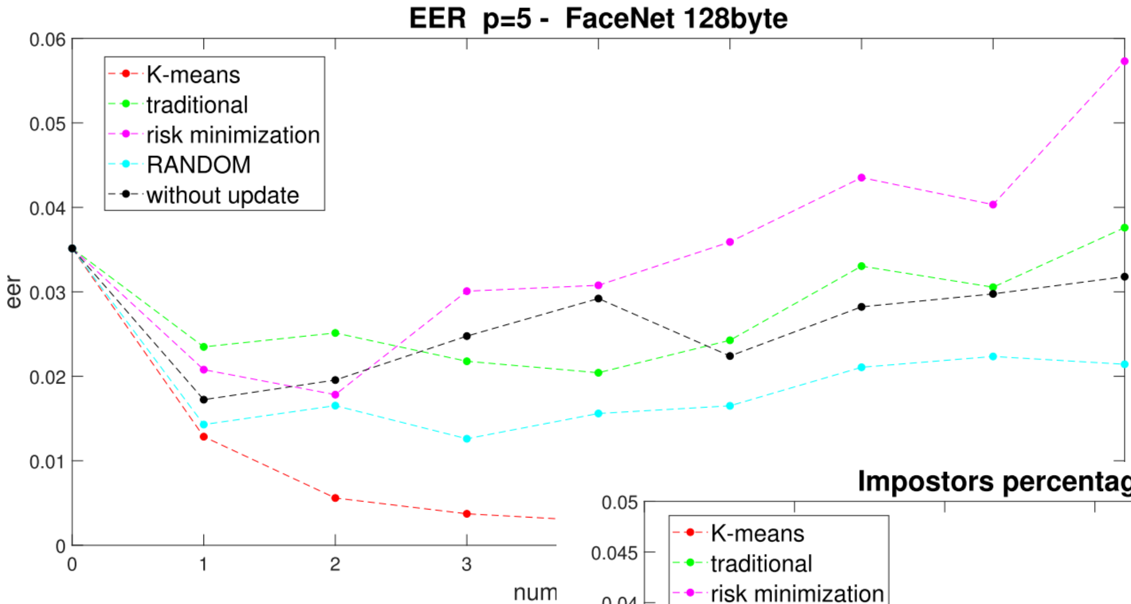
https://www.saiferlab.ai/#

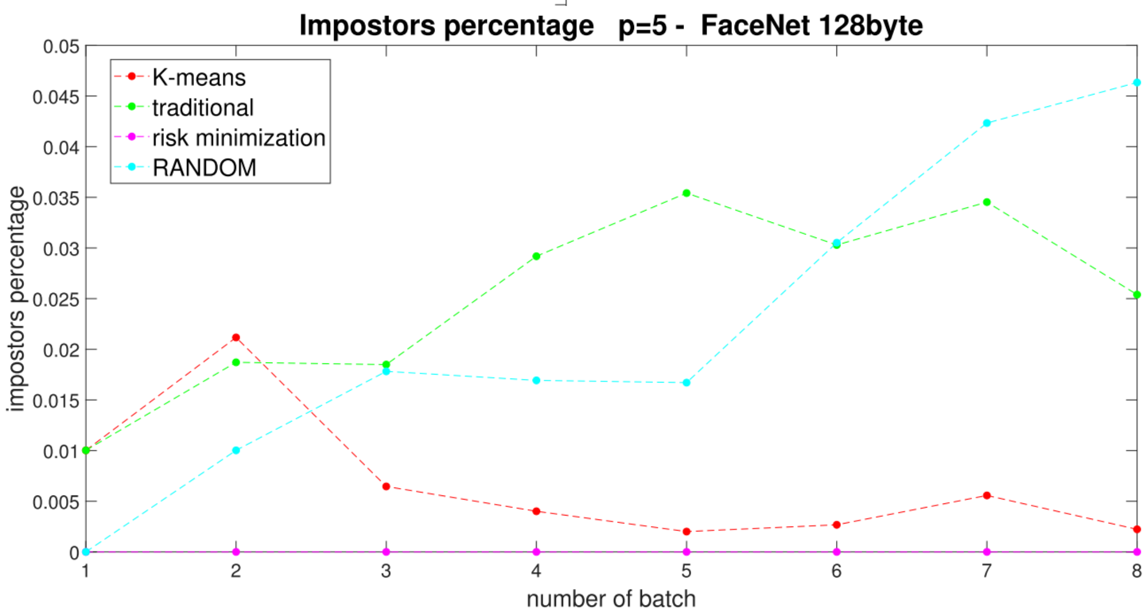
Adaptive systems parameters



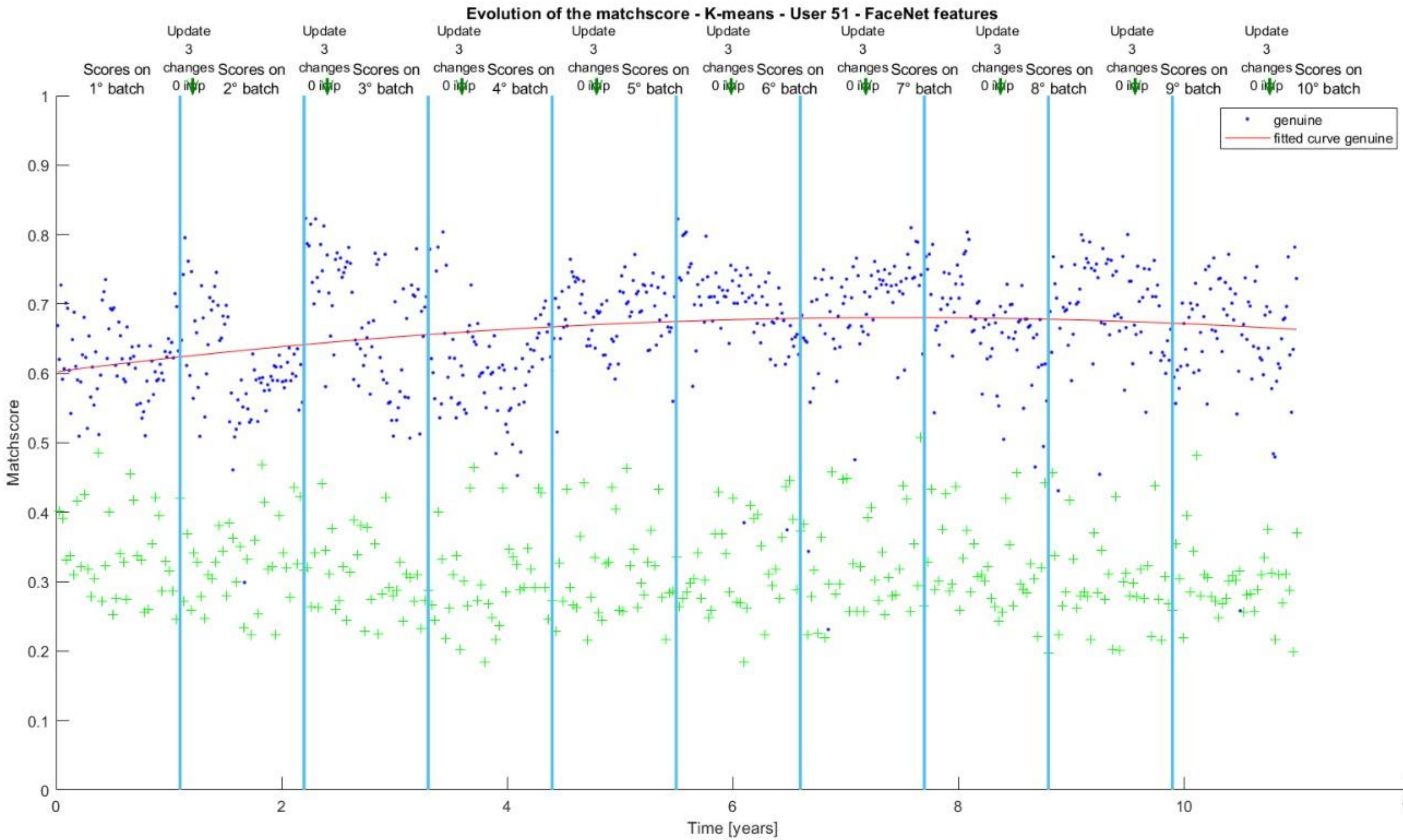
Samples added to the clients galleries may impact on the modules of the system

Some experiments…





Effect on users over time



User templates over updating

Initial Template Gallery

Template Gallery after 1° update cycle

Template Gallery after 2° update cycle

Template Gallery after 3° update cycle

Template Gallery after 4° update cycle

Template Gallery after 5° update cycle

Template Gallery after 6° update cycle

Template Gallery after 7° update cycle

Template Gallery after 8° update cycle

Template Gallery after 9° update cycle

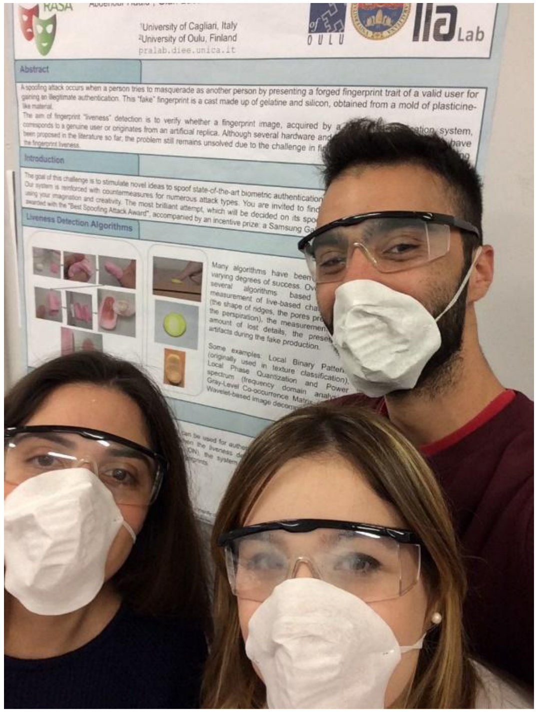
Homework

- **Design a PCA-based adaptive biometric systems**
  - Without changing the eigenvectors once trained first time
  - By changing the eigenvectors for each samples accepted into a user's gallery
  - By changing the eigenvectors and selecting the best p templates according to criterion like:
    - Last Recently Used template substitution
    - Minimum average distance among possible templates (MDIST)
- **Plot accuracy and rank curves, store the users galleries**

That’s all for today… what’s next?

- **Multi-modal biometric systems**
  - properties
  - pros and cons
- **Fusion approaches**
  - Feature-level fusion
  - Measurement-level fusion
  - Decision-level modern challenge
- **Fusion architectures**
  - Parallel
  - Serial

Thank you for listening!



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